Mathematical Glossary for `Analysis/FabiusFunction`

Encyclopedic definitions for the Lean development and its documentation

Terminology synthesized from Vladimir Reshetnikov's `ProveIt` repository

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Abstract

This glossary inventories the mathematical terminology used throughout the Lean comments and module documentation under `Analysis/FabiusFunction`, together with the mathematical vocabulary appearing in the associated Markdown and LaTeX documents. It contains 437 alphabetized headwords with detailed conceptual definitions, characteristic formulas, and explanations of how each notion functions inside the Fabius/Rvachev development. The typography, page geometry, headings, colors, and running-head design follow the repository article `docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`. Elementary words such as "number," "function," "sum," and "interval" are omitted unless a specialized compound gives them a technical meaning.

Editorial unit of inclusion. The inventory is concept-based rather than token-based. Inflectional and spelling variants are consolidated, while a familiar alias may retain a short entry when the repository uses it to emphasize a different formula or operational role; such an entry explicitly identifies the canonical concept rather than inventing a distinction. A named repository object receives its own entry when its normalization matters, as for the bounded Fabius function, the signed global extension, the sequences c_n, d_n, F_n, G_n, R_n, the lower Lambert coordinate, and the periodic correction Ψ. Lean implementation vocabulary is included only when it denotes a mathematical representation or construction, such as a fast signed-dyadic name or the predicate `IsFabius`.
Definitions here are explanatory rather than substitutes for theorem statements. Whenever a domain, endpoint convention, coercion, normalization, or convergence mode is material, the Lean declaration named in the formalization crosswalk is authoritative. In particular, sharing a traditional name does not identify two differently normalized repository objects.

Object and notation guardrails. Unless a definition says otherwise, F is the bounded CDF-style function, $\mathcal{F}$ is the signed global extension that agrees with it on $[0, 1]$, and up is Rvachev's even compactly supported function. The Fourier convention is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi i x\xi}\, \mathrm{d}x$. Binary weight is written $\mathrm{wt}_2(n)$, and the dyadic valuation is v_2. Empty sums are zero, empty products are one, and natural-number exponentiation uses $0^0 = 1$ in the order-zero cases of the finite identities.

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Headword index

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Absolute continuity. A function $f : [a, b] \to \mathbb{R}$ is absolutely continuous if, for every $\varepsilon > 0$, some $\delta > 0$ makes $\sum_j |f(b_j) - f(a_j)| < \varepsilon$ whenever the intervals (a_j, b_j) are pairwise disjoint and have total length below δ. Equivalently, f has an integrable derivative almost everywhere and $f(x) = f(a) + \int_a^x f'(t)\ \mathrm{d}t$. A measure μ is absolutely continuous with respect to Lebesgue measure if every Lebesgue-null set is μ-null; the Radon–Nikodym theorem then represents μ by a density. These two meanings meet in the Fabius CDF, whose derivative is a smooth compactly supported density.

Absolute convergence. A series $\sum_{n \geq 0} a_n$ in $\mathbb{R}$ or $\mathbb{C}$ is absolutely convergent when $\sum_{n \geq 0} |a_n| < \infty$. Absolute convergence is stronger than ordinary convergence and permits rearrangement, regrouping, and many exchanges of limits with finite algebraic operations. For a series of functions, pointwise absolute convergence is distinguished from uniform absolute convergence, which is usually established by a summable majorant. In this development the signed global Fabius series and the complex q-binomial series are proved absolutely convergent, so their Thue–Morse signs do not hide merely conditional cancellation.

Admissible class. An admissible class is the collection of objects on which a defining operator is intended to act and within which existence or uniqueness is asserted. Conditions may include boundedness, continuity, prescribed endpoint values, symmetry, monotonicity, support restrictions, or differentiability. For the bounded Fabius function, a convenient admissible class consists of continuous maps with constant tails and the correct reflection law; the refinement or integration operator preserves this class and is contractive in the supremum norm. Stating the class explicitly prevents a local functional equation from being mistaken for a global characterization.

Affine transformation. An affine transformation of one real variable has the form $x \mapsto ax + b$, with $a \neq 0$; it combines scaling, reflection when $a < 0$, and translation. Affine changes of variable transport support intervals, densities, derivatives, and Fourier transforms in controlled ways. The relations $\mathrm{up}(x) = F(1 - |x|)$, $F(x) = \mathrm{up}(x - 1)$ on $[0, 1]$, and the derivative identity $F'(x) = 2\,\mathrm{up}(2x - 1)$ are built from such transformations. Dyadic self-similarity uses the special dilation $x \mapsto 2x$.

All-orders asymptotic expansion. An all-orders asymptotic expansion is a hierarchy of finite approximations valid to arbitrary prescribed order. Writing $f(x) \sim \sum_{j \geq 0} a_j \phi_j(x)$ in an asymptotic scale means that for every N, subtracting the first N terms leaves $o(\phi_{N-1}(x))$, or a stated sharper remainder. It does not generally assert convergence of the infinite formal series. For the Fabius endpoint, the repository proves expansions in inverse powers of the lower-Lambert coordinate $\lambda(x)$, with smooth one-periodic coefficient functions that encode the surviving dyadic oscillation.

Almost everywhere. A property holds almost everywhere (a.e.) with respect to a measure μ if the set where it fails is μ-null. Functions equal a.e. have the same Lebesgue integral and represent the same element of an L^p space, but they need not have the same pointwise continuity or endpoint values. Lean packages this notion in the almost-everywhere filter and in restricted measures. The

Fabius measure-theoretic bridge uses a.e. identities for interval kernels and integrals, while the final differential and support identities concern canonical continuous or smooth representatives and therefore hold pointwise.

Almost-sure convergence. Random variables X_n converge almost surely to X if $\mathbb{P}(\{\omega : X_n(\omega) \to X(\omega)\}) = 1$. This pathwise mode implies convergence in probability. In the Fabius model, $X = \sum_{k\geq 1} 2^{-k} U_k$ with $0 \leq U_k \leq 1$, so the series converges for every outcome by comparison with $\sum_{k\geq 1} 2^{-k} = 1$, a statement stronger than almost-sure convergence. Almost-sure convergence constructs the limiting random variable; weak convergence or characteristic functions then identify its law as the Fabius distribution.

Amplitude. In an integral treated by the saddle-point or steepest-descent method, the amplitude is the factor multiplying the dominant exponential phase, as in $I(\lambda) = \int A(z, \lambda) \exp(\lambda\Phi(z))\,\mathrm{d}z$. The phase determines the saddle and Gaussian localization, while derivatives of the amplitude contribute to correction coefficients. In the Fabius Bromwich analysis, factors originating in the Laplace product, the Jacobian of the local change of variables, and reciprocal powers of the saddle coordinate form the amplitude. Separating amplitude from phase organizes the all-orders coefficient algebra.

Analytic continuation. Analytic continuation extends a holomorphic function from one domain to a larger connected domain by requiring agreement on their overlap. Uniqueness follows from the identity theorem: two holomorphic continuations that agree on a set with an accumulation point agree everywhere on the connected component. Mellin transforms and zeta expressions in the endpoint analysis are first defined in regions of absolute convergence and then continued meromorphically. The procedure must be distinguished from merely assigning a formal value to a divergent expression; poles and finite parts have to be tracked explicitly.

Analytic germ. The analytic germ of a function at a point records its behavior on some unspecified neighborhood of that point, with two representatives identified when they agree on a smaller neighborhood. A smooth function has an analytic germ at x_0 precisely when it equals a convergent power series near x_0. At dyadic interior points the Fabius formal Taylor series terminates, but the function does not agree locally with that polynomial; hence no analytic germ is present there. This distinction is central to proving nowhere analyticity despite exact finite Taylor data.

Analytic locus. The analytic locus of a smooth real function is the set of points at which it is real analytic on some neighborhood. It is open by definition. A constant tail is analytic at every point strictly outside the transition interval, whereas flatness at a boundary point need not imply analyticity there. The repository identifies the exact analytic loci of the bounded Fabius function, the compactly supported up-function, and the signed global extension, carefully separating exterior constant regions, support endpoints, and the dense set of dyadic points where Taylor series terminate without representing the function.

Analytic moment. An analytic moment is a moment regarded not only as an integral but as a Taylor coefficient of an analytic transform. If $M(z) = \int \mathrm{e}^{zx}\,\mu(\mathrm{d}x)$ is entire, then $m_n = M^{(n)}(0)$. This viewpoint gives Cauchy estimates, generating-function identities, and scalar extension to complex arguments. The repository's analytic-moment modules connect the exact Fabius moment recurrences with the entire complex moment-generating function.

Approximate identity. An approximate identity is a family of kernels K_n whose mass is normalized, whose mass concentrates near the origin, and whose convolution with a suitable function tends to that function. In the Fabius probability construction, the omitted-tail laws concentrate at the origin and hence form an approximate identity. Convolving a finite partial-sum law with its omitted-tail law recovers the full law. The finite spline laws themselves converge to the Fabius law and are not an approximate identity.

Approximation theory. Approximation theory studies how functions are represented or approximated by simpler classes such as polynomials, splines, rational functions, or translates of a fixed kernel. It asks for convergence, rates, optimality, stability, and constructive algorithms. The up-function belongs to Rvachev's atomic-function approach, while the repository develops both centered spline approximations with explicit uniform error and Fourier-Legendre polynomial approximations with least-squares optimality.

Asymptotic equivalence. For nonzero comparison functions, $f(x) \sim g(x)$ as $x \to a$ means $f(x)/g(x) \to 1$. This multiplicative statement is stronger than equality of leading logarithmic orders. If $\log f = \log g + o(1)$, then asymptotic equivalence follows after exponentiation; an $O(1)$ logarithmic discrepancy generally does not suffice. In the Fabius endpoint problem, the periodic correction must be included in the exponential main term: omitting a nonconstant bounded function changes the ratio by a nonconvergent oscillatory factor and destroys the claimed equivalence.

Asymptotic expansion. An asymptotic expansion expresses a function by successively smaller comparison terms near a limiting regime. If $\phi_{j+1} = o(\phi_j)$, the notation $f \sim \sum_{j\geq 0} a_j \phi_j$ means that every finite truncation approximates f to the scale of its last retained term. Coefficients may be constants, polynomials, or periodic functions. The Fabius development uses logarithmic expansions, saddle expansions, and inverse-Lambert expansions; their remainders are stated quantitatively rather than inferred from a formal infinite series.

Asymptotic scale. An asymptotic scale is an ordered family (ϕ_j) with $\phi_{j+1}(x) = o(\phi_j(x))$ in the limit under study. It supplies the notion of successive order needed for an asymptotic expansion. Powers of $1/\lambda$, where $\lambda \to \infty$, form the main scale in the corrected Fabius saddle analysis; powers of $1/|\log x|$ are a coarser equivalent scale after expanding the lower Lambert coordinate. Logarithmic and log-periodic factors can modify the scale and must not be silently absorbed into constants.

Atomic function. In Rvachev's terminology, an atomic function is a compactly supported smooth solution of a functional-differential or refinement equation from which localized approximation systems can be generated. The canonical example is the up-function, supported on $[-1, 1]$ and satisfying $\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big)$. It combines compact support, C^∞-smoothness, exact self-similarity, and polynomial-reproduction identities, yet is nowhere analytic on its support. Here "atomic" means a fundamental building block; it is unrelated to an atom of a measure.

Automatic sequence. A sequence is k-automatic if a finite automaton reading the base-k digits of n outputs its n-th term. The Thue–Morse sequence is the fundamental 2-automatic example: the automaton tracks the parity of the number of one bits. Automatic structure yields digit recurrences, finite block factorization, and efficient generation. These properties drive the Fabius signed translate and dyadic cancellation formulas.

Averaging operator. An averaging operator replaces a value by an integral or finite mean over translated or rescaled values. A typical probability operator is $(Tf)(x) = \int f(x-y)\,\mu(\,\mathrm{d}y)$, namely convolution with a probability measure. Such operators preserve positivity and constants, contract the supremum norm, and smooth irregularities. The fixed-point construction of the Fabius law can be formulated with an operator induced by a scaled uniform random variable; iterating it corresponds to adding independent dyadic-scale summands and produces finite spline approximants.

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B-spline. A B-spline is a compactly supported piecewise polynomial obtained from repeated convolution of interval indicators, with degree and smoothness increasing under convolution. Uniform B-splines use equally spaced knots; centered forms are symmetric about the origin. Finite convolutions of the dyadically scaled uniform densities in the probabilistic Fabius construction are nonuniformly scaled box splines, and after a common-grid reformulation they yield explicit centered finite splines. These approximants have rational dyadic data, controlled support, and a certified uniform error.

Banach fixed-point theorem. The Banach fixed-point theorem states that a strict contraction T on a nonempty complete metric space has a unique fixed point x_*. If $d(Tx, Ty) \le qd(x, y)$ with $0 < q < 1$, then $d(T^n x_0, x_*) \le q^n d(Tx_0, x_0)/(1-q)$. In the Fabius construction, the ambient space is a closed class of bounded continuous functions with the supremum metric, and the defining integral operator is contractive. This proves existence, uniqueness, and a certified approximation scheme before smoothness is known; differentiating the fixed-point equation then bootstraps the fixed point to C^∞.

Basic hypergeometric series. A basic, or q-hypergeometric, series is a power series whose ratio of consecutive coefficients is a rational function of q^n, rather than of n as in an ordinary hypergeometric series. It is built from q-Pochhammer symbols $(a;q)_n$. Finite Gaussian binomial identities are terminating examples. The Fabius dyadic formulas use the special base $q = 1/2$, where finite products translate into odd Mersenne factors and interpolation weights; no convergence issue arises for terminating identities, while the global presentations require separate absolute-convergence proofs.

Bell polynomial. Bell polynomials encode the combinatorics of differentiating a composite exponential or logarithm. The complete exponential Bell polynomial satisfies $\exp(\sum_{m\ge1} x_m t^m/m!) = \sum_{n\ge0} B_n(x_1,\ldots,x_n)t^n/n!$; partial Bell polynomials refine the number of factors. They provide a compact form of the Faà di Bruno formula. In saddle expansions, they convert phase and amplitude jets into coefficients of the exponential series, while logarithmic coefficients can also be expressed through ordered compositions. The repository often spells out these finite sums to facilitate formal verification.

Bernoulli number. The Bernoulli numbers B_n are rational numbers defined by $t/(\mathrm{e}^t - 1) = \sum_{n \geq 0} B_n t^n/n!$, with the convention $B_1 = -1/2$. They govern sums of powers, Euler–Maclaurin summation, and the Taylor expansions of trigonometric and Bose-type kernels. The Fabius arithmetic development derives Bernoulli recurrences related to moment sequences and reciprocal-sine products. Because two sign conventions for B_1 occur in the literature, a formal development must fix the generating function rather than rely on the name alone.

Bernoulli recurrence. A Bernoulli recurrence is a triangular identity determining Bernoulli numbers from earlier values, classically $\sum_{k=0}^{n} \binom{n+1}{k} B_k = 0$ for $n \geq 1$. Equivalent recurrences arise by multiplying or differentiating the generating function $t/(\mathrm{e}^t - 1)$. In the repository, Bernoulli-type relations connect coefficients in reciprocal trigonometric expansions with Fabius moment recurrences. Triangularity means the newest coefficient appears with a nonzero scalar and can therefore be solved exactly over the rationals.

Bessel inequality. Bessel's inequality states that for an orthonormal system (e_n) in a Hilbert space, $\sum |\langle f, e_n \rangle|^2 \leq \|f\|^2$. Equality holds for a complete basis and becomes Parseval's identity. Applied to normalized Legendre polynomials, it bounds coefficient squares and the L^2 error of partial sums. Stronger coefficient estimates are needed for the repository's absolute uniform convergence theorem.

Big-O and little-o notation. The relation $f(x) = \mathcal{O}(g(x))$ means that $|f(x)| \leq C|g(x)|$ near the specified limit for some constant C; $f(x) = o(g(x))$ means $f(x)/g(x) \to 0$. Uniform qualifiers indicate that the constant or convergence is independent of auxiliary variables. These symbols describe error scales, not hidden equalities. In the corrected Fabius asymptotic, an $\mathcal{O}(1/\lambda(x))$ logarithmic remainder is meaningful only after the nonconstant periodic main term has been retained.

Bijection. A bijection is a map that is both injective and surjective, so every target element has exactly one preimage. A continuous strictly monotone function from an interval onto an interval is a bijection and admits a continuous inverse. The bounded Fabius function is constant outside $[0, 1]$, but its restriction to $[0, 1]$ is a strictly increasing bijection onto $[0, 1]$. This fact licenses the inverse Fabius function: for a dyadic ordinate one may ask for the unique abscissa having that value. These are inverse-function questions at dyadic ordinates, not the repository's "inverse-dyadic values," which are samples at inverse powers of two.

Bilateral Laplace transform. The bilateral, or two-sided, Laplace transform is $\mathcal{L}_b f(s) = \int_{-\infty}^{\infty} \mathrm{e}^{-sx} f(x) \, \mathrm{d}x$ on the vertical strip where the integral converges. It restricts to a Fourier transform on the imaginary axis and, for a compactly supported density, extends to an entire function. Centered Fabius transforms factor into elementary uniform-law factors. Their positive-real Laplace specialization—equivalently the negative-real specialization of the moment-generating convention—produces the negative-Laplace product used in Bromwich inversion and the endpoint saddle analysis. The qualifier distinguishes it from the usual one-sided transform on $[0, \infty)$.

Binary digit sum. The binary digit sum $\mathrm{wt}_2(n)$, also called Hamming weight or popcount, is the number of ones in the base-two expansion of n. Its parity generates the Thue–Morse sign $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}$, and $\mathrm{wt}_2(2n) = \mathrm{wt}_2(n)$, $\mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n) + 1$ yield the fundamental block recursions. Lucas's theorem also shows that row n of Pascal's triangle contains exactly $2^{\mathrm{wt}_2(n)}$ odd entries. Thus the same statistic controls Thue–Morse cancellation, parity of the Fabius numerator sequences, exact 2-adic valuations, and the number of recursive descents in the highest-set-bit evaluator.

Binary expansion. A binary expansion represents a nonnegative real number as $x = \sum_j b_j 2^j$ or, on the unit interval, as $x = \sum_{j\geq 1} b_j 2^{-j}$ with digits $b_j \in \{0, 1\}$. Dyadic rationals have two infinite expansions, one terminating and one ending in infinitely many ones, so algorithms must impose a convention or prove representation independence. The global binary reduction in the repository peels off scale blocks according to the set bits and produces an absolutely convergent series whose value is independent of harmless encoding choices.

Binary reduction series. A binary reduction series is a scale-by-scale expansion obtained by iterating an identity that reduces the argument according to its binary digits. In the Fabius setting, a Taylor block is extracted at each active binary scale, and the residual argument is shifted and rescaled. The finite reductions telescope; passing to the limit yields an absolutely convergent representation of the signed global extension. Unlike a mere digit algorithm, the construction is an analytic identity and includes explicit coefficients, convergence estimates, and a proof that the series agrees with the bounded Fabius function on $[0, 1]$.

Binomial coefficient. The binomial coefficient $\binom{n}{k} = n!/(k!(n-k)!)$ counts k-element subsets of an n-element set and is the coefficient of x^k in $(1 + x)^n$. Pascal's recurrence and the binomial theorem make it ubiquitous in derivative scaling, moment recurrences, Taylor expansions, and finite differences. The Fabius formulas combine ordinary binomial coefficients with Gaussian q-binomial coefficients; keeping the two notations distinct is essential because the latter depend on a base parameter and encode geometric rather than equally spaced nodes.

Binomial transform. The binomial transform sends a sequence (a_k) to $b_n = \sum_{k=0}^{n} \binom{n}{k} a_k$, with signed variants obtained by inserting $(-1)^k$. It is inverted by the corresponding signed transform. Such transforms arise when moments are shifted, when an exponential generating function is multiplied by e^z, or when a variable is replaced by $1 - x$. In the Fabius arithmetic, the full moments and half moments are connected by finite even-index binomial sums produced by the symmetry and translation between the bounded and centered forms.

Borel probability measure. A Borel probability measure assigns total mass one to the sigma-algebra generated by open sets. On $\mathbb{R}$, such a measure is Radon: it is regular and finite on compact sets. It is determined by its cumulative distribution function and, when absolutely continuous, by a density. The law of $\sum_{k\geq 1} 2^{-k} U_k$ for independent $U_k \sim \mathrm{Unif}[0, 1]$ is a compactly supported Borel probability measure. Its CDF is the bounded Fabius function, while the centered affine image $2X - 1$ has density equal to the Rvachev up-function.

Bose-Einstein kernel. The logarithmic Bose–Einstein kernel used in this repository is $B(x) = \log(1 - \mathrm{e}^{-x})$ for $x > 0$. Expanding $B(x) = -\sum_{m\ge1} \mathrm{e}^{-mx}/m$ and integrating termwise gives the Mellin transform

$$\int_0^\infty x^{a-1} B(x)\ \mathrm{d}x = -\Gamma(a)\zeta(a+1), \qquad a > 0.$$

Its logarithmic singularity at zero produces a double Mellin pole after continuation. A finite-part subtraction isolates the constant term, while sampling the continued Gamma–zeta expression on the dyadic imaginary lattice yields the Fourier coefficients of the periodic endpoint correction.

Bose finite-part integral. A Bose finite-part integral is a regularized integral involving a Bose-Einstein kernel whose endpoint singular terms have been explicitly removed. If an integrand has an expansion $a_{-m}t^{-m} + \cdots + a_{-1}t^{-1} + O(1)$, one subtracts the singular part, integrates the remainder, and adds the canonically determined finite contributions. In the repository this construction supplies constants and periodic corrections in the Mellin analysis of the negative-Laplace product. It is a Hadamard finite part, not an ordinary improper integral when the unsubtracted integral diverges.

Boundary value problem. A boundary value problem seeks a solution of a differential or functional-differential equation satisfying conditions at boundary points or on the boundary of a domain. Compact support can be viewed as an infinite collection of exterior boundary conditions, while endpoint flatness enforces smooth gluing to zero. Rvachev atomic functions were developed for nonclassical approximation methods in boundary value problems because their localized dilates and translates can encode domain geometry.

Bounded Fabius function. The bounded Fabius function is the normalization $F : \mathbb{R} \to [0, 1]$ that equals zero for $x \le 0$, one for $x \ge 1$, satisfies $F(1 - x) = 1 - F(x)$, and obeys $F'(x) = 2F(2x)$ on $[0, 1/2]$. The adjective bounded distinguishes it from the signed global extension $\mathcal{F}$, which satisfies the pure dilation equation everywhere but is not a CDF. It is the normalization used for dyadic values, inverse values, and computability.

Bounded variation. A real or complex function on an interval has bounded variation if the supremum of $\sum_j |f(x_j) - f(x_{j-1})|$ over all partitions is finite. For a finite row of weights, total variation means $\sum_j |\omega_{n,j}|$. Uniform bounded variation prevents cancellation from concealing unbounded coefficients and is the stability hypothesis in a Toeplitz convergence argument. The complex q-binomial discrete limit uses rows whose mass is one and whose total variation is uniformly bounded by an explicit finite product ratio.

Bromwich inversion integral. The Bromwich inversion integral recovers a function from a Laplace transform along a vertical line. In the repository's endpoint regime, Fourier inversion yields for every $r > 0$ the exact representation

$$F(x) = \mathrm{e}^{rx} \int_{\mathbb{R}} \mathrm{e}^{2\pi\mathrm{i}\xi x} \frac{G(-(r + 2\pi\mathrm{i}\xi))}{r + 2\pi\mathrm{i}\xi}\ \mathrm{d}\xi,$$

where G is the relevant generating product. Thus the contour coordinate $z = r+2\pi\mathrm{i}\xi$ has positive real part while the product is sampled at $-z$. The proof derives this line directly rather than presupposing a contour shift; analytic decay then supports the quantitative saddle estimates.

Bromwich saddle method. The Bromwich saddle method applies steepest-descent ideas directly to a Laplace inversion contour. Classically one identifies a large real saddle solving the derivative equation for the logarithmic phase, then deforms or localizes the vertical contour, expands the phase, and bounds tails and minor arcs. In the Fabius analysis the reference balance and radius are expressed through the lower branch W_{-1}; residual phase terms are retained and bounded rather than erased by calling that reference point the full critical point. The method yields the main exponential scale, log-periodic correction, Gaussian mass, and systematic inverse-coordinate corrections.

Bump function. A bump function is a C^∞ function with compact support. Nonzero bump functions cannot be real analytic on all of $\mathbb{R}$, because an analytic function vanishing on an open set vanishes identically. Rvachev's up-function is a distinguished bump: it is even, nonnegative, supported on $[-1, 1]$, strictly unimodal, infinitely flat at ± 1, and satisfies an exact refinement-type differential equation. Its special algebraic structure makes it useful far beyond the generic existence of smooth cutoffs.

3 C

C-infinity smoothness. A function is C^∞ if derivatives of every finite order exist and are continuous. Smoothness is weaker than real analyticity: the Taylor series may fail to converge to the function, or may even be identically zero at a flat point. The Fabius and up-functions are canonical examples of smooth but nowhere-analytic behavior on their nonconstant regions. Their differential scaling law gives exact formulas and sharp supremum bounds for all derivatives, while compact support forces infinite-order vanishing at the boundary.

Canonical solution. A canonical solution is a solution selected uniquely by natural normalization and regularity conditions, rather than by an arbitrary parameter choice. A differential or functional equation alone may admit many local or distributional solutions; support, symmetry, endpoint values, positivity, and smoothness can isolate one. The bounded Fabius function is canonical as the unique smooth distribution function with the stated tails, symmetry, and dyadic differential equation. Rvachev's up-function is the corresponding canonical compactly supported even density.

Cauchy integral formula. If f is holomorphic inside and on a positively oriented simple closed contour γ, then $f^{(n)}(z_0) = \frac{n!}{2\pi \mathrm{i}} \int_\gamma f(z)/(z - z_0)^{n+1} \, \mathrm{d}z$. It implies analyticity, derivative bounds, and coefficient extraction. Transform functions in the Fabius development are entire, so Cauchy estimates can control Taylor coefficients and justify analytic scalar manipulations.

Cauchy product. The Cauchy product of $\sum a_n z^n$ and $\sum b_n z^n$ is $\sum_{n \geq 0} (\sum_{k=0}^{n} a_k b_{n-k}) z^n$. Under absolute convergence it represents the product of the sums; for formal power series it is the definition of multiplication without analytic convergence. Moment and Bernoulli recurrences arise by equating Cauchy-product coefficients in generating-function identities. In formalized proofs, finite coefficient sums are preferable to informal series multiplication because every index range and normalization is explicit.

Cauchy theorem. Cauchy's theorem says that the integral of a holomorphic function around a closed contour is zero under standard domain hypotheses. Consequently, integrals along homotopic contours agree unless singularities are crossed. It underlies the deformation of Bromwich and Mellin contours used in the endpoint analysis. Quantitative tail bounds are still required because the contours are unbounded.

Centered finite spline. The centered finite spline named in the computability layer is globally a signed Thue–Morse spline. On $[0,1]$, and at positive degree, it agrees with the CDF of a finite centered, dyadically weighted sum of uniform variables; its derivative on polynomial pieces is the corresponding spline density. It is piecewise polynomial on a dyadic grid and converges uniformly to the signed extension globally, hence to the bounded Fabius function on the unit interval, with an explicit tail error. Its dyadic values are computable by natural-number operations. The degree-zero step case needs half-weighted endpoints, and the centering correction prevents a systematic half-mesh displacement.

Centered periodic correction. A centered periodic correction is a periodic function from which its mean value has been subtracted. If $P(u+1)=P(u)$, then $\Psi(u)=P(u)-\int_0^1 P(v)\,\mathrm{d}v$ has period one and zero average. Centering moves the constant Fourier mode into the nonoscillatory main constant and leaves only nonzero modes. In the corrected Fabius endpoint asymptotic, $\Psi(\lambda(x))$ is a genuine nonconstant log-periodic term whose Fourier coefficients are explicit Gamma–zeta values.

Characteristic function. The characteristic function of a real random variable X is $\varphi_X(t)=\mathbb{E}(\mathrm{e}^{\mathrm{i}tX})$. It always exists, is positive definite and uniformly continuous, and determines the probability law uniquely. Independence turns sums into products: $\varphi_{X+Y}=\varphi_X\varphi_Y$. For the centered dyadically weighted uniform series underlying the up-function, each summand contributes a sinc factor, so the characteristic function becomes an infinite sinc product. Under the repository's Fourier convention this is precisely the Fourier transform of the density.

Closed form. A closed form is a finite expression assembled from an agreed collection of standard operations or special functions. The term has no universally fixed boundary: a finite sum involving moments and Thue–Morse signs may be considered explicit even when no elementary simplification exists. In the Fabius development, closed forms include exact dyadic finite sums, explicit recurrences, Gaussian q-binomial formulas, and Lambert-based asymptotic coordinates. The phrase should not imply that evaluation is numerically stable or computationally optimal.

Coefficient extraction. Coefficient extraction is denoted $[z^n]A(z)$ and returns the coefficient of z^n in a polynomial, formal power series, or convergent analytic series. It converts product identities into finite convolution sums and turns interpolation or generating-function statements into arithmetic formulas. In the q-binomial part of the Fabius theory, a weighted Thue–Morse block annihilates all lower-degree node polynomials, leaving a leading coefficient that equals the desired dyadic value after normalization.

Compact support. The support of a function is the closure of the set where it is nonzero; compact support means this closed set is compact. On $\mathbb{R}$, such a function vanishes outside a bounded interval. The up-function has support exactly $[-1, 1]$. Compact support implies that its Fourier transform extends to an entire function of exponential type, while C^∞ smoothness and endpoint flatness give rapid decay along the real axis. Support is topological: endpoints belong to it even though the function value there is zero.

Complex exponential generating function. An exponential generating function has the form $A(z) = \sum_{n \geq 0} a_n z^n / n!$. When $z \in \mathbb{C}$ and the series converges, it is a complex analytic function; formally, it packages binomial convolutions especially efficiently. Moment generating functions are exponential generating functions of moments. For the Fabius and up distributions, complex EGFs connect the moment recurrences with products of elementary exponential factors and permit coefficient identities to be proved uniformly over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

Complex moment-generating function. For a bounded real random variable X, the complex moment-generating function is $M_X(z) = \mathbb{E}(\mathrm{e}^{zX})$ for $z \in \mathbb{C}$. Boundedness makes it entire, and $M_X^{(n)}(0) = \mathbb{E}(X^n)$. Independence converts sums into products. Evaluating at imaginary arguments gives the characteristic function, while negative real arguments lead to Laplace asymptotics. The Fabius development uses this single entire object to relate moments, the sinc product, the negative-Laplace product, and Bromwich inversion.

Complex shift. A complex shift replaces an argument or interpolation node by $z + Q$ with a fixed $Q \in \mathbb{C}$. Translation usually changes finite expressions, but a sufficiently high-order finite-difference functional annihilates all lower-degree contributions introduced by the shift. The repository proves that each normalized finite q-binomial Taylor block is independent of a common complex shift, even though individual summands depend on it. In the associated discrete-limit approximants, dependence may remain at finite level and disappears only in the limit.

Computable real. A computable real is a real number supplied by an algorithm that, given a precision request p, returns a rational or signed dyadic approximation within 2^{-p}. The algorithm must be uniform and terminating, and the error guarantee is part of the representation. A computable sequence requires one algorithm uniform in both the sequence index and the precision. The repository uses redundant pairs of natural-number numerators to keep the coding primitive recursive and to avoid relying on hidden integer arithmetic.

Computable real function. In the Grzegorczyk-style formulation used here, a real function is computable when it is sequentially computable and effectively uniformly continuous. Sequential computability means that it maps every computable real sequence to a computable sequence, uniformly in indices and requested precision. Effective uniform continuity supplies a recursive modulus controlling how accurate the input must be. The bounded Fabius function is proved computable by evaluating explicit finite splines and using a certified uniform approximation error together with the sharp global Lipschitz bound.

Computable sequence. A sequence $(x_i)_{i\geq 0}$ of real or complex numbers is computable if a single algorithm, given i and a precision parameter p, produces a rational approximation to x_i with a certified error such as 2^{-p}. Uniformity in i distinguishes a computable sequence from a list of individually computable numbers with unrelated algorithms. Moment sequences, dyadic values, and coefficients of spline approximants in the repository are represented by exact recurrences, yielding stronger information than approximation alone.

Conditional asymptotic theorem. A conditional asymptotic theorem derives a limiting formula under an explicit hypothesis not yet proved in the surrounding development, such as a parity assertion, a tail estimate, or a nonvanishing condition. Its logical status is weaker than an unconditional theorem even when the derivation is correct. Several repository modules retain conditional intermediate results as reusable transfer lemmas, while later modules discharge the hypotheses. The glossary distinguishes this proof-theoretic adjective from conditional convergence of series, which is a different analytic notion.

Contour deformation. Contour deformation replaces an integration path in the complex plane by a homologous path on which the integral is easier to estimate, using Cauchy's theorem and accounting for any crossed singularities by residues. For an unbounded contour one must also control connecting arcs and convergence at infinity. In Bromwich saddle analysis, the original vertical line is localized or deformed toward a steepest-descent path through the saddle. The repository separates local central estimates from tail and minor-arc bounds so that the deformation is quantitatively justified.

Contour integral. A contour integral $\int_\gamma f(z)\,\mathrm{d}z$ integrates a complex function along a parameterized curve γ. Holomorphy allows path deformation, while poles contribute residues. Fourier inversion, Laplace inversion, Mellin inversion, and coefficient extraction all admit contour formulations. In the Fabius endpoint proof, the contour integral carries a large exponential phase determined by the negative-Laplace transform; its central portion yields the saddle expansion and its remote portions require uniform decay estimates.

Contraction constant. A contraction constant is a number $q < 1$ satisfying $d(Tx, Ty) \leq qd(x, y)$. It governs the geometric rate q^n of fixed-point iteration and the size of certified error bounds. In a Fabius fixed-point construction, the exact value depends on the chosen operator and function norm. Recording it turns uniqueness into an effective approximation theorem.

Contraction mapping. A contraction mapping on a metric space satisfies $d(Tx, Ty) \leq qd(x, y)$ for a constant $q < 1$. Contractions are automatically continuous and, on complete spaces, have unique fixed points. Their iterates converge geometrically, giving explicit numerical error bounds. The Fabius defining operator is arranged to be a contraction on a space of bounded continuous functions with fixed tails and symmetry. This construction avoids assuming beforehand that the desired smooth function or probability law exists.

Convexity and concavity. A function is convex on an interval when $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$ for $0 \leq t \leq 1$; it is concave when the inequality is reversed. For twice differentiable functions, $f'' \geq 0$ implies convexity and $f'' \leq 0$ implies concavity. The bounded Fabius function is strictly convex on the left half of $[0,1]$, strictly concave on the right half, and has a unique inflection point at $1/2$. This follows from the monotonicity and symmetry of its derivative, which is an affine copy of the up-function.

Convolution. For integrable functions, $(f*g)(x)=\int_{\mathbb{R}} f(x-y)g(y)\,\mathrm{d}y$; for measures, convolution is the law of the sum of independent random variables. Convolution is associative and commutative, preserves total mass, and turns into multiplication under the Fourier transform. In general $\operatorname{supp}(f*g)\subseteq\overline{\operatorname{supp}f+\operatorname{supp}g}$; for compact supports the closure is unnecessary, and equality still needs positivity or nondegeneracy to exclude cancellation. The Fabius probability model is an infinite convolution of scaled uniform laws. Finite convolutions are splines, while the limiting centered density is the up-function.

Convolution of measures. For finite Borel measures μ,ν on $\mathbb{R}$, their convolution is $(\mu*\nu)(A)=\int\int \mathbf{1}_A(x+y)\,\mu(\mathrm{d}x)\nu(\mathrm{d}y)$. It is the pushforward of $\mu\otimes\nu$ under addition. Characteristic functions multiply, and the convolution support is contained in the closure of the Minkowski sum; equality holds for the positive compact interval laws used here. The finite and infinite convolution measures in the Fabius probability construction are defined at this measure level before smooth densities are identified.

Convolution power. The k-fold convolution power f^{*k} is defined recursively by $f^{*1}=f$ and $f^{*(k+1)}=f^{*k}*f$. It represents the density of a sum of k independent identically distributed variables when f is a density. Refinement identities for the up-function lead to exact self-convolution and scaled convolution formulas. Fourier transformation converts a convolution power into an ordinary power of the transform, often making normalization and support calculations immediate.

Correction term. A correction term is a subordinate contribution added to a leading approximation to reach a sharper error scale. Corrections can be algebraic powers, logarithms, periodic functions, or exponentially small terms. Their relative size must be judged in the variable actually governing the limit. In the Fabius endpoint expansion, the centered one-periodic fluctuation is part of the main exponent rather than a dispensable numerical decoration; subsequent saddle corrections occur in inverse powers of $\lambda(x)$.

Cosine series. A cosine series is a Fourier series involving only $\cos(n\pi x/L)$, appropriate for even functions on a symmetric interval. If f is even, its sine coefficients vanish. The up-function admits a cosine expansion whose coefficients are samples of its Fourier transform; the sinc product forces all nonzero even-frequency coefficients in a particular normalization to vanish, leaving an odd cosine series. Smoothness gives rapid coefficient decay and hence strong convergence.

Cumulant. Cumulants κ_n are the coefficients of the logarithm of a moment-generating or characteristic function: $\log\mathbb{E}(\mathrm{e}^{zX})=\sum_{n\geq 1}\kappa_n z^n/n!$ near zero. The first cumulant is the mean, the second is the variance, and higher cumulants measure non-Gaussian structure. Cumulants add for independent sums, making them natural for the infinite random-series representation. Endpoint saddle modules use tilted cumulants and their recurrences to control local phase derivatives and sharp dyadic asymptotics.

Cumulant-generating function. The cumulant-generating function is $K_X(z) = \log M_X(z)$, where $M_X(z) = \mathbb{E}(e^{zX})$, on a domain where a consistent logarithm is defined. Its derivatives at zero are cumulants. For products of moment-generating functions arising from independent summands, taking logarithms converts the product into a sum, revealing dyadic harmonic structure. The negative-real specialization for the Fabius law is denoted by a logarithmic Laplace function whose exact decomposition contains polynomial-logarithmic terms and a one-periodic remainder.

Cumulative distribution function. The cumulative distribution function of a real random variable X is $F(x) = \mathbb{P}(X \leq x)$. It is nondecreasing, right-continuous, tends to zero at $-\infty$, and tends to one at $+\infty$. If the law has a continuous density f, then $F'(x) = f(x)$. The bounded Fabius function is a smooth CDF supported in the probabilistic sense on $[0, 1]$; its derivative is a scaled translate of Rvachev's up-function.

4 D

Decay estimate. A decay estimate bounds a function by a quantity tending to zero in a specified limit, often uniformly in auxiliary parameters. Polynomial, exponential, superpolynomial, and stretched-exponential decay have different analytic consequences. The Fourier transform of a compactly supported C^∞ function decays faster than every power on the real axis, while the Fabius endpoint value decays on a quadratic logarithmic scale. Quantitative decay estimates justify Fourier inversion, contour truncation, termwise operations, and numerical stopping rules.

Delay differential equation. A delay differential equation relates a derivative at one point to function values at transformed or earlier points, rather than only to values at the same point. The Fabius relation $F'(x) = 2F(2x)$ is more precisely an advanced dilation equation on the left half when read in x, but after logarithmic change of variables it becomes a delay equation because multiplication of x by two becomes translation. Such equations encode self-similarity and can generate log-periodic phenomena.

Denominator bound. A denominator bound gives an explicit integer divisible by the reduced denominators of a family of rational quantities. It may be universal over all dyadic numerators at a fixed level or tailored to one sequence. In the Fabius arithmetic, exact finite formulas prove rationality at dyadic points, while valuation formulas identify powers of two and least-common-multiple products control odd prime factors. A denominator bound certifies exact storage and comparison even when it is not minimal.

Derivative scaling. Derivative scaling is the law obtained by repeatedly differentiating a dilation equation. From $\mathcal{F}'(x) = 2\mathcal{F}(2x)$, induction and the chain rule give $\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x)$. The triangular exponent comes from multiplying $2^{1+2+\cdots+m}$. This identity yields exact dyadic derivatives, sharp uniform derivative bounds, terminating Taylor series at dyadic points, and high-order flatness estimates.

Differential equation. A differential equation imposes a relation among a function and its derivatives. Ordinary differential equations compare quantities at the same independent variable, while functional-differential equations may include dilated or translated arguments. The Fabius and up equations are of the latter type. Their global solution theory requires normalization and support or tail conditions; without these, the equation alone need not select the intended function. Repeated differentiation reveals the exact self-similar hierarchy of derivatives.

Differentiation under the integral sign. Differentiation under the integral sign exchanges $\frac{\mathrm{d}}{\mathrm{d}x}\int f(x,t)\ \mathrm{d}t$ with $\int \partial_x f(x,t)\ \mathrm{d}t$. It is justified by hypotheses such as continuity of the derivative and an integrable dominating function uniform in x. The method derives smoothness and differential identities for convolution and transform representations. In the Fabius setting, compact support and rapid decay often provide the needed domination, but endpoint or complex-contour integrals require separate estimates.

Digitwise factorization. Digitwise factorization is an identity in which a sum indexed by binary words factors into independent contributions from each bit position. The basic Thue–Morse polynomial satisfies $\sum_{r=0}^{2^k-1} \varepsilon_r z^r = \prod_{j=0}^{k-1}(1-z^{2^j})$. Choosing a bit either contributes 1 or contributes $-z^{2^j}$, so expanding the product enumerates all binary integers and records the parity of their digit sum. This factorization is the engine behind Prouhet cancellation and several dyadic formulas.

Dirac measure. The Dirac measure δ_a assigns mass one to every measurable set containing a and zero otherwise; integration evaluates the integrand, $\int f\ \mathrm{d}\delta_a = f(a)$. Finite weighted sums of Dirac measures form the discrete polynomial measures in the early Fabius approximants, convolution combines independent discrete coordinates, and δ_0 is the convolution unit. The weak limit becomes a continuous compactly supported law. A Dirac measure is not an ordinary "delta function"; its Fourier transform under the repository convention is the phase $\mathrm{e}^{-2\pi \mathrm{i} a\xi}$.

Dirichlet eta function. The Dirichlet eta function is $\eta(s) = \sum_{n\geq 1}(-1)^{n-1}n^{-s}$ for $\Re s > 0$, with analytic continuation to an entire function. It satisfies $\eta(s) = (1-2^{1-s})\zeta(s)$. Alternating sums involving powers or reciprocal powers can therefore be expressed in zeta terms with a dyadic correction factor. Eta-type combinations appear naturally when Thue–Morse or parity decompositions separate even and odd indices.

Discrete B-spline. A discrete B-spline is a finite sequence or grid function obtained by repeated discrete convolution of interval indicators, serving as the lattice analogue of a continuous B-spline. Iterated prefix sums are an equivalent construction. After normalization and rescaling, such sequences approximate compactly supported spline densities. The repository identifies corrected Thue–Morse grid arrays with histogram or discrete-spline data, allowing a combinatorial recurrence to be connected to the analytic up-function through a discrete limit.

Discrete convolution. For sequences a, b, their discrete convolution is $(a * b)_n = \sum_k a_k b_{n-k}$, with the sum restricted to indices where both terms are defined. It models the distribution of sums of integer-valued random variables and is the coefficient rule for products of generating functions. Iterated convolution of finite box sequences produces discrete B-splines. In Thue–Morse limit arguments, discrete convolution and prefix summation reveal cancellation and smooth the signed data before scaling.

Discrete limit. A discrete limit theorem shows that normalized finite arrays, sums, or grid functions converge to a continuous object as the mesh tends to zero or the degree tends to infinity. Convergence may be pointwise, uniform, in L^p, or weak. The Fabius directory contains several discrete-limit constructions: spline grids converging to the up-function, complex-shift q-binomial rows converging to the signed global extension, and Thue–Morse histograms converging after exact normalization. Finite identities and stability bounds are kept separate from the limiting argument.

Distribution law. The distribution law of a random variable is the probability measure it induces on its value space. Two random variables can be defined on different probability spaces yet have the same law. Laws are compared through cumulative distribution functions, moments when determinate, or characteristic functions. The Fabius function is most naturally defined as the CDF of an infinite weighted sum of independent uniform variables; this law-based viewpoint makes positivity, symmetry, and convolution identities immediate.

Division-free recurrence. A division-free recurrence computes each new term using additions and multiplications of previously known integers, with any apparent rational factors already shown integral. Such a recurrence is valuable for exact arithmetic because it avoids repeated fraction reduction and exposes positivity or divisibility. The repository gives division-free recurrences for normalized moment numerators and denominator sequences by clearing triangular factors such as factorials and Mersenne products. The proof must verify that every displayed quotient is an integer before the recurrence can be called division-free.

Dominated convergence theorem. The dominated convergence theorem states that if measurable functions $f_n \to f$ almost everywhere and $|f_n| \le g$ for an integrable g, then $\int f_n \to \int f$. It is a principal tool for interchanging limits with integrals. In the Fabius theory it supports limits of finite convolutions, differentiation or expansion of transform integrals, and passage from finite random sums to the infinite law. Uniform or local domination must be checked explicitly when a parameter varies.

Double factorial. The double factorial is $n!! = n(n-2)(n-4)\cdots$, terminating at 1 or 2, with $(-1)!! = 0!! = 1$. Thus $(2m-1)!! = (2m)!/(2^m m!)$. Odd double factorials are Gaussian moments and appear in saddle-point integrals, the normalization of the full-moment sequence c_n, and the common denominator bound B_n. Their product structure isolates powers of two from odd factors, which is useful in valuation and integrality arguments.

Dyadic argument. A dyadic argument is a real input of the form $a/2^n$, usually with integers $a, n \ge 0$. Reducing a to odd numerator identifies the exact dyadic level. The Fabius differential scaling maps dyadic points to integers after finitely many doublings, causing all sufficiently high derivatives to vanish there. Consequently the formal Taylor series terminates and exact rational evaluation becomes possible by finite recurrences or Thue–Morse sums.

Dyadic common denominator. For $n \ge 0$, the repository's canonical D_n is the least common multiple of the reduced denominators of $\mathrm{up}(a/2^n)$ over positive odd numerators $1 \le a < 2^n$. By reflection, it equals the separately exposed LCM of the denominators of $F(a/2^n)$ on the same odd grid. Thus D_n uniformly clears an entire dyadic level. The denominator theorem proves $D_n \mid B_n$, where

$$B_n = n!\, 2^{\binom{n+1}{2}}(1 \cdot 3 \cdots (2\lfloor n/2 \rfloor + 1)) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1).$$

This D_n must not be conflated with moment numerators called C_n, D_n in some source conventions.

Dyadic grid. The dyadic grid of level n is $2^{-n}\mathbb{Z}$, or its intersection with a bounded interval. It is nested under refinement: every level-n point also belongs to level $n+1$. Functional equations with dilation by two interact especially well with this grid. Fabius values and derivatives on it admit finite exact formulas, while spline approximants and numerical verification suites use the same grid for certified comparison.

Dyadic harmonic sum. A dyadic harmonic sum has the form $H(s) = \sum_{j \in \mathbb{Z} \text{ or } j \ge 0} h(s/2^j)$. Mellin transformation turns scaling by 2^{-j} into a geometric series factor $(1 - 2^{-z})^{-1}$. Its imaginary lattice of poles produces one-periodic functions of $\log_2 s$ or a related phase. The logarithm of the negative-Laplace Fabius product is such a sum.

Dyadic name. A dyadic name for a real number is a sequence of dyadic rationals $a_p/2^p$ converging with a prescribed fast error, commonly $|x - a_p/2^p| \le 2^{-p}$. A signed-dyadic name may encode the numerator as a pair of naturals (a_p^+, a_p^-). Names are representations used in computable analysis: an algorithm operates on approximations with guarantees rather than on an inaccessible exact real. The Fabius computability proof constructs such names uniformly from finite spline values.

Dyadic rational. A dyadic rational is a rational number whose reduced denominator is a power of two, equivalently an integer multiple of 2^{-n}. Dyadic rationals form a dense subring of $\mathbb{R}$ and are exactly the finite binary fractions. They are privileged for the Fabius function because the dilation equation repeatedly doubles the argument; after finitely many steps a dyadic point reaches an integer, producing terminating derivative and evaluation formulas. Every bounded Fabius value at a dyadic argument is rational.

Dyadic refinement invariance. Dyadic refinement invariance is the property that representing the same dyadic rational on a finer grid does not change the value produced by a formula or algorithm. For example, $a/2^n = (2a)/2^{n+1}$ must yield identical outputs. This is nontrivial when formulas contain level-dependent moments, factorials, or sign sums. The repository proves compatibility directly, ensuring that exact evaluators depend on the rational number itself rather than on a noncanonical numerator-denominator pair.

Dyadic Taylor evaluator. The dyadic Taylor evaluator is a terminating exact algorithm for $F(a/2^n)$. It uses a Taylor identity whose derivatives at a dyadic center are known through the global scaling law and vanish above a finite order. A binary recursion removes a highest set bit or folds the numerator into a smaller block, so the number of recursive calls is tied to the binary weight rather than to the denominator size. All arithmetic is rational and the termination proof is explicit.

Dyadic valuation. The dyadic, or 2-adic, valuation $v_2(r)$ of a nonzero rational r is the exponent of two in its prime factorization: write $r = 2^k u/v$ with odd integers u, v, then $v_2(r) = k$. It satisfies $v_2(rs) = v_2(r) + v_2(s)$ and $v_2(r+s) \geq \min(v_2(r), v_2(s))$. For $n \geq 1$, the repository proves

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!),$$

thereby determining the power of two in the reduced denominator; the case $n = 0$ is excluded because $F(1) = 1$.

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5 E

Effective flatness. Effective flatness is a quantitative version of infinite-order vanishing at an endpoint. Instead of merely asserting $F^{(m)}(0) = 0$ for every m, it gives explicit bounds such as $F(x) \leq 2^{\binom{n+1}{2}} x^n/n!$ when $2^n x \leq 1$. Such inequalities provide computable rates, justify endpoint truncations, and show decay faster than every fixed power. The repository derives both mean-value and iterated-integral forms from sharp derivative scaling.

Effective modulus. An effective modulus converts a qualitative continuity or convergence statement into a recursive rate. For uniform continuity, a function $d(n)$ may guarantee that $|x-y| < 1/d(n)$ implies $|f(x) - f(y)| < 1/n$. For convergence, a modulus gives an index after which an error tolerance is met. The bounded Fabius function has an explicit modulus derived from its least Lipschitz constant, while the spline approximation has a direct precision-to-level rule. These data are essential in computable analysis.

Effective uniform continuity. A function is effectively uniformly continuous when it has a recursively computable modulus valid over its whole domain. In the convention used by the repository, a recursive positive integer $d(n)$ satisfies $|x-y| < 1/d(n) \Rightarrow |f(x)-f(y)| < 1/n$ for every positive n. This strengthens ordinary uniform continuity by exposing a usable rate. For the constant-tail bounded Fabius function, the global Lipschitz bound yields the primitive-recursive choice $d(n) = 2n$.

Elementary logarithmic asymptotic. An elementary logarithmic asymptotic rewrites a special-function coordinate, such as the lower Lambert variable, in nested elementary logarithms. For small x, $W_{-1}(-cx)$ can be expanded using $\log x$, $\log|\log x|$, and inverse powers of $|\log x|$. Substitution makes the scale more familiar but also lengthens the formula and can obscure the periodic phase. The repository derives this form only after establishing the exact Lambert-coordinate expansion, so error propagation remains controlled.

Empty sum, empty product, and zero-power conventions. Neutral-element conventions make finite identities valid at order zero: an empty sum is 0, an empty product is 1, the unique composition of 0 is the empty composition, and natural-number exponentiation uses $0^0 = 1$. These choices determine the zeroth rows of the moment, q-binomial, composition, and binary-reduction formulas. They are mathematical boundary data, not cosmetic notation: starting a sum at one or discarding an empty term can make an otherwise correct identity fail only at an endpoint, which is particularly easy to miss in informal derivations.

Endpoint asymptotic. An endpoint asymptotic describes a function as its argument approaches the boundary of its support or domain, here $x \downarrow 0$ for the bounded Fabius function. Infinite flatness makes ordinary Taylor expansion useless: every derivative at the endpoint vanishes. The correct analysis instead uses Laplace transforms, a saddle whose location diverges, a lower-Lambert coordinate, and a log-periodic correction. Symmetry transports the result to the opposite endpoint.

Endpoint flatness. A smooth function is flat relative to its endpoint value at a if all derivatives of positive order vanish there. Taylor's theorem then gives $f(x) - f(a) = o(|x-a|^N)$ for every fixed N; only when $f(a) = 0$ may this be shortened to $f(x) = o(|x-a|^N)$. Thus F is flat at zero, $1 - F$ is flat at one, and up is flat at both support endpoints. Flatness is necessary for smooth extension by a constant tail but does not imply analyticity. The repository also supplies effective bounds quantifying this infinite-order vanishing.

Entire function. An entire function is holomorphic on all of $\mathbb{C}$. Its Taylor series at any point has infinite radius of convergence, though its growth can vary greatly. The Fourier transform of the compactly supported up-function is entire and is represented by a locally uniformly convergent infinite product of sinc factors. Its zeros, growth, and real-axis decay encode partitions of unity, sampling identities, and inversion formulas. Entirety follows from the product or from compact support together with standard Fourier-Laplace theory.

Entire function of exponential type. An entire function F is of exponential type τ if roughly $|F(z)| \le C_\varepsilon \exp((\tau + \varepsilon)|z|)$ for every $\varepsilon > 0$. Fourier transforms of functions supported in a compact interval have exponential type controlled by the support radius. For the up-function supported on $[-1, 1]$, this growth is compatible with rapid decay on the real axis. The distinction between complex-plane growth and real-axis decay is important: an entire function can grow exponentially off the real axis while decreasing superpolynomially along it.

Equicontinuity. A family $\mathcal{F}$ of functions is equicontinuous if one neighborhood size works simultaneously for every member of the family at each point; uniform equicontinuity uses one size over the whole domain. Equicontinuity, boundedness, and compactness interact through the Arzela–Ascoli theorem. In approximation arguments for finite Fabius splines or convolutions, a common Lipschitz bound gives equicontinuity and helps upgrade pointwise or weak convergence to uniform convergence on compact sets.

Euler-Maclaurin formula. The Euler–Maclaurin formula connects sums and integrals by endpoint derivatives and Bernoulli numbers. In one form, $\sum_{n=a}^{b} f(n) = \int_a^b f(t)\, \mathrm{d}t + (f(a) + f(b))/2 +$

$\sum_{k\geq 1} B_{2k}(f^{(2k-1)}(b) - f^{(2k-1)}(a))/(2k)! + R$. It is a principal tool for extracting constants, zeta values, and asymptotic corrections from harmonic sums. The Fabius Mellin and saddle analysis uses Euler–Maclaurin-type evaluations with explicit remainder control rather than as a purely formal expansion.

Euler-Mascheroni constant. The Euler–Mascheroni constant is $\gamma = \lim_{n\to\infty}(\sum_{k=1}^{n} 1/k - \log n)$. It is the constant term in the Laurent expansion $\zeta(s) = 1/(s-1) + \gamma + \cdots$ and satisfies $\Gamma'(1) = -\gamma$. Finite-part Mellin calculations and expansions of Gamma-zeta products naturally produce γ. In an endpoint asymptotic, it belongs to the nonoscillatory constant after the periodic component has been centered.

Exact arithmetic. Exact arithmetic manipulates integers, rationals, algebraic data, or symbolic products without rounding error. It contrasts with floating-point evaluation, where cancellation and underflow can erase tiny endpoint values. The dyadic Fabius algorithms use integer signs, binomial coefficients, moment numerators, factorials, and known denominators to return exact rational results. These values serve both as mathematical theorems and as high-precision reference data for validating asymptotic or numerical procedures.

Exact dyadic derivative. If $0 < a < 2^n$, a is odd, and $x = a/2^n$, then

$$F^{(n)}(x) = 2^{\binom{n+1}{2}}(-1)^{\mathrm{wt}_2((a-1)/2)}, \qquad F^{(m)}(x) = 0 \quad (m > n).$$

This follows from global derivative scaling because the signed extension satisfies $\mathcal{F}(2b+1) = (-1)^{\mathrm{wt}_2(b)}$ and vanishes at even integers. Thus the terminating Taylor polynomial at a reduced interior dyadic point has degree exactly n, with a fully specified leading sign.

Expectation. The expectation $\mathbb{E}[X]$ of an integrable random variable is its probability-weighted mean, defined as a Lebesgue integral. More generally, $\mathbb{E}[g(X)]$ integrates g against the law of X. Moments, characteristic functions, and moment-generating functions are expectations of X^n, $\mathrm{e}^{\mathrm{i}tX}$, and e^{zX}. In the Fabius probability representation, linearity of expectation and independence translate the infinite weighted sum into exact moment recurrences and transform products.

Explicit uniform error. An explicit uniform error bound has the form $\sup_{x\in D}|f_n(x) - f(x)| \leq E_n$ with a fully specified $E_n \to 0$. It is stronger than an existence statement and directly yields a stopping rule. The centered finite-spline approximation to the bounded Fabius function satisfies a dyadic error bound independent of x. This estimate is used both for certified numerical evaluation and for constructing a computable real function with a recursive precision modulus.

Exponential generating function. For a sequence (a_n), the exponential generating function is $A(z) = \sum_{n\geq 0} a_n z^n/n!$. The factorial denominator makes multiplication encode binomial convolution: the coefficient of $z^n/n!$ in $A(z)B(z)$ is $\sum_k \binom{n}{k} a_k b_{n-k}$. Moment-generating functions are canonical examples. Fabius moment and half-moment recurrences become compact product or functional equations when expressed in this form.

Extended nonnegative reals. The extended nonnegative reals $\mathbb{R}_{\geq 0} \cup \{+\infty\}$, written $\mathbb{R}_{\geq 0\infty}$ or `ENNReal` in Lean, are the natural codomain for measures and infinite sums of nonnegative quantities. Probability measures ultimately have finite mass one, but intermediate measure identities and monotone limits are most naturally stated before finiteness is proved. The early Fabius measure bridge therefore establishes identities in `ENNReal`, proves the relevant values are not infinite, and only then applies `toReal` to obtain ordinary real-valued CDF and integral statements.

Extension by zero. Extension by zero takes a function defined on a subset and declares it to vanish outside that subset. For a function on a closed interval to extend as a C^m function, its derivatives through order m must match zero at the boundary; C^∞ extension therefore requires flatness. Rvachev's up-function is naturally a compactly supported global function, while finite spline pieces are often first defined on their support and then extended by zero. Endpoint normalization determines whether the extension has the claimed regularity.

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6 F

Faa di Bruno formula. The Faa di Bruno formula is the higher-order chain rule for a composite $f \circ g$. One form is $\frac{\mathrm{d}^n}{\mathrm{d}x^n} f(g(x)) = \sum_{k=1}^{n} f^{(k)}(g(x)) B_{n,k}(g'(x), \ldots, g^{(n-k+1)}(x))$, where $B_{n,k}$ are partial Bell polynomials. It organizes the derivatives of exponentials, logarithms, and inverse functions. In the all-orders saddle and inverse-Lambert algebra, the formula explains why coefficients are finite polynomials in lower-order phase jets, even when the Lean proof unfolds them through recurrences.

Fabius distribution. The Fabius distribution is the compactly supported probability law whose cumulative distribution function is the bounded Fabius function. It can be realized by an infinite sum of independent uniform random variables with geometrically decreasing weights, after the normalization used in the repository. Its law is symmetric about $1/2$, has a smooth density on $(0, 1)$, and is infinitely flat at the support endpoints. Centering and rescaling the density yields Rvachev's up-function.

Fabius function. The bounded Fabius function is the unique C^∞ map $F : \mathbb{R} \to [0, 1]$ with $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, reflection law $F(1 - x) = 1 - F(x)$, and differential equation $F'(x) = 2F(2x)$ on $[0, 1/2]$. It is a strictly increasing distribution function on $[0, 1]$, strictly convex then strictly concave, and nowhere analytic on its nonconstant interval. Its dyadic values are rational, and its endpoint decay contains a nonconstant log-periodic correction.

Fabius recurrence sequence. A Fabius recurrence sequence is one of the exact rational or integer sequences introduced to encode moments, inverse-power values, dyadic numerators, or normalized endpoint values. Such sequences are usually triangular: the n-th term is determined from earlier terms by binomial coefficients, factorials, and factors $2^j - 1$. The phrase is repository-specific rather than universally standardized, so the defining recurrence and normalization must accompany it. Different source papers use related symbols for moments, half moments, numerators, and Reshetnikov integers.

Fast signed-dyadic name. A fast signed-dyadic name represents a real number at precision p by a pair $(c^+, c^-) \in \mathbb{N}^2$, interpreted as $(c^+ - c^-)/2^p$, with error at most 2^{-p}. The adjective fast refers to the guaranteed exponential error rate in the precision index, not necessarily to computational complexity. This redundant coding avoids negative integers inside primitive-recursive constructions and is extensionally equivalent to ordinary signed dyadic approximation. It is the representation used in the repository's computable-analysis layer.

Filter convergence. A filter abstracts which sets count as "eventually" in a limiting process. A function f tends to y along a filter $\mathcal{F}$ when the inverse image of every neighborhood of y belongs to $\mathcal{F}$. Sequences use the cofinite filter on $\mathbb{N}$; $x \to 0^+$ uses a right-neighborhood filter; almost-everywhere statements use the measure filter. Lean formulates continuity, series, and asymptotic relations through this uniform language. Making the filter explicit prevents the direction and domain of the Fabius endpoint, lower-Lambert, and contour limits from being silently changed.

Finite difference. The forward finite-difference operator is $(\Delta_h f)(x) = f(x+h) - f(x)$; its n-fold iterate is an alternating binomial sum. It annihilates polynomials of degree below n and extracts the leading coefficient of a degree-n polynomial up to $n!h^n$. Gaussian q-differences replace equally spaced nodes by geometric nodes and ordinary binomial weights by q-binomial weights. The Fabius dyadic formulas combine such leading-coefficient extraction with Thue–Morse cancellation.

Finite-part integral. A finite-part integral assigns a canonical finite value to an integral with a controlled endpoint divergence by subtracting the singular asymptotic terms before taking the limit. For example, if $f(t) = a_{-1}/t + O(1)$, the finite part of $\int_0^1 f(t)\ \mathrm{d}t$ removes $a_{-1}\log(1/\varepsilon)$ from the cutoff integral. It differs from a Cauchy principal value, which relies on symmetric cancellation. Mellin and Bose-kernel calculations in the Fabius asymptotic use Hadamard finite parts to isolate constants and periodic terms.

Finite spline approximation. A finite spline approximation replaces the limiting Fabius law by the distribution of a finite partial random sum, whose density is a compactly supported piecewise polynomial. Every approximant can be evaluated using finite rational arithmetic on a dyadic grid. Centering the support removes a half-step bias, and coupling the omitted tail gives an explicit uniform error. These approximants are simultaneously probabilistic truncations, convolution splines, and constructive witnesses for computability.

First saddle correction. The first saddle correction is the coefficient of $1/\lambda$ after the leading Gaussian saddle approximation. It combines the quartic phase term, the square of the cubic phase term, derivatives of the amplitude, and any first-order periodic dependence of the transform. In logarithmic form it becomes $A_1(\lambda(x))/\lambda(x)$. The repository derives it explicitly and proves an error one order smaller.

Fixed point. A fixed point of a map T is an element x satisfying $T(x) = x$. Functional equations often become fixed-point equations on spaces of functions. A unique fixed point can be obtained from contraction, order, compactness, or monotone-iteration arguments. The Fabius distribution is constructed as the fixed point of an averaging/refinement operator; the operator identity is equivalent to the probabilistic self-similarity of the underlying random series and ultimately to the differential equation.

Flat function. A smooth function is flat at a point x_0 if $f^{(n)}(x_0) = 0$ for every $n \geq 0$, or, in some conventions, for every positive n after the value is separated. The model e^{-1/x^2} extended by zero is flat but not analytic. The Fabius function is flat at its transition endpoints relative to its constant tails, and the up-function is flat at ± 1. Effective flatness estimates quantify the otherwise qualitative vanishing of every derivative.

Formal logarithm. For a formal power series $A(z) = 1 + U(z)$ with zero constant term in U, its formal logarithm is $\log A = \sum_{r \geq 1}(-1)^{r-1}U^r/r$. Every coefficient depends on only finitely many coefficients of U, so no analytic convergence is required. In the saddle expansion, the normalized mass series is converted into logarithmic correction coefficients by this operation. Ordered compositions give an explicit coefficient formula for the formal logarithm.

Formal power series. A formal power series $\sum_{n \geq 0} a_n z^n$ is an algebraic object whose coefficients are manipulated without assigning a numerical value to z or requiring convergence. Addition is coefficientwise and multiplication is the Cauchy product. Formal differentiation, composition under suitable constant-term conditions, exponentiation, and logarithms are defined coefficientwise. The repository uses formal series to certify moment recurrences, saddle jets, and all-orders coefficient identities before separate analytic estimates justify any asymptotic interpretation.

Fourier coefficient. For a one-periodic integrable function P, its k-th complex Fourier coefficient is $\widehat{P}(k) = \int_0^1 P(u)\mathrm{e}^{-2\pi \mathrm{i} k u}\,\mathrm{d}u$. The coefficient $k = 0$ is the mean; nonzero coefficients measure oscillation. In the corrected Fabius endpoint theory, the centered periodic fluctuation has coefficients proportional to $-\Gamma(-\chi_k)\zeta(1-\chi_k)/\log 2$, where $\chi_k = 2\pi \mathrm{i} k/\log 2$. Their nonvanishing proves that the fluctuation is genuinely nonconstant.

Fourier inversion. Fourier inversion reconstructs a function from its Fourier transform under suitable integrability or regularity assumptions. With the convention $\widehat{f}(\xi) = \int f(x)\mathrm{e}^{-2\pi \mathrm{i} x \xi}\,\mathrm{d}x$, one has $f(x) = \int \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x \xi}\,\mathrm{d}\xi$ when both sides are appropriately integrable, with standard variants at discontinuities. The up-function's rapid transform decay makes inversion straightforward and supports cosine expansions, derivative formulas, and sampling identities.

Fourier-Legendre coefficient. The Fourier-Legendre coefficient of $f \in L^2[-1,1]$ is the scalar projection onto a Legendre polynomial: $a_n = (2n+1)\int_{-1}^1 f(x)P_n(x)\,\mathrm{d}x/2$ in the standard normalization. These coefficients minimize mean-square error among polynomial approximations when truncated. The repository derives exact coefficients for the up-function, exploiting moments and symmetry, and proves more than L^2 convergence: the resulting series converges absolutely and uniformly.

Fourier-Legendre series. A Fourier-Legendre series expands a function on $[-1,1]$ in the orthogonal basis (P_n) of Legendre polynomials. It is analogous to a trigonometric Fourier series but adapted to polynomial approximation with unit weight. For even functions only even degrees occur. The up-function's series has exact arithmetic coefficients, least-squares optimality of partial sums, and absolute uniform convergence established through derivative and coefficient estimates. It should not be confused with the ordinary Fourier transform of the same function.

Fourier mode. A Fourier mode is a basic exponential $u \mapsto \mathrm{e}^{2\pi \mathrm{i} k u}$ or its sine-cosine components at integer frequency k. Periodic functions decompose into such modes. Under a dyadic dilation equation, complex exponents with imaginary parts $2\pi k/\log 2$ recur because multiplication by two advances the logarithmic phase by one period. The nonzero modes of the Fabius periodic correction are expressed by Gamma–zeta coefficients and cannot be replaced by a constant.

Fourier series. A Fourier series represents a periodic function as $\sum_{k\in\mathbb{Z}} c_k \mathrm{e}^{2\pi \mathrm{i} k x}$, or equivalently by sine and cosine terms. Convergence may be in L^2, pointwise, uniform, or absolute, depending on regularity. Smooth periodic functions have rapidly decaying coefficients. The repository uses Fourier series both for the endpoint's one-periodic correction and for periodicizations of the up-function; the two applications have different variables and should not be conflated.

Fourier transform. The Fourier transform converts translation into multiplication by a phase, convolution into multiplication, and differentiation into multiplication by frequency. Under the repository's convention, $\widehat{f}(z) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} x z}\,\mathrm{d}x$. For the up-function it equals the entire product $\prod_{n\geq 0} \operatorname{sinc}(\pi z/2^n)$. Zeros of the first factors yield partition and sampling identities, while compact support and smoothness determine complex growth and real-axis decay.

Fourier-transform convention. A Fourier-transform convention specifies the signs and factors of 2π in the transform and inverse transform. The reference article uses $\widehat{f}(z) = \int f(x)\mathrm{e}^{-2\pi \mathrm{i} x z}\,\mathrm{d}x$, for which the centered interval density has a $\operatorname{sinc}(\pi z)$ transform. Sampling frequencies, derivative multipliers, and Poisson-summation constants all depend on this choice. Stating the convention prevents normalization errors.

Fractional part. The floor $\lfloor x \rfloor$ is the greatest integer at most x, the ceiling $\lceil x \rceil$ is the least integer at least x, and the fractional part is $\{x\} = x - \lfloor x \rfloor \in [0, 1)$. Any one-periodic function $P(x)$ can be viewed as a function of $\{x\}$. Floors and ceilings also make finite dyadic ranges and endpoint cases exact. In endpoint asymptotics the relevant phase is the fractional part of a logarithmic or Lambert coordinate; the periodic correction records the residual dependence after an approximate integer shift under dyadic rescaling.

Fubini theorem. Fubini's theorem permits the order of integration in a product space to be exchanged when the integrand is absolutely integrable. Tonelli's theorem covers nonnegative measurable integrands without prior integrability. These results justify factorization of expectations for independent variables, convolution formulas, and rearrangement of transform integrals. Signed Thue–Morse expressions require absolute bounds before Fubini can be invoked.

Full logarithmic expansion. The full logarithmic expansion is the all-orders expansion of $\log F(x)$, rather than of $F(x)$ itself. In the repository it has the form $\log F(x) =$

$\mathcal{M}(x) + \sum_{j=1}^{N-1} A_j(\lambda(x))/\lambda(x)^j + \mathcal{O}(\lambda(x)^{-N})$, where the A_j are smooth one-periodic functions. Taking the logarithm separates multiplicative saddle corrections into additive coefficients and makes the uniqueness and size of the periodic term transparent.

Full moment sequence c_n. The repository's full moment sequence is

$$c_n = \int_{\mathbb{R}} t^{2n}\ \mathrm{up}(t)\ \mathrm{d}t = \frac{F_n}{(2n+1)!!\prod_{j=1}^{n}(2^{2j}-1)}.$$

Evenness of up makes the omitted odd moments zero. The rational recurrence for c_n feeds the dyadic closed formula and Fourier–Legendre coefficients, while the integer numerator F_n exposes parity and divisibility. This normalization is fixed by the displayed integral; the symbol F_n here denotes a sequence, not the function F.

Functional-differential equation. A functional-differential equation combines derivatives with values at transformed arguments, for example $f'(x) = 2f(2x)$ or an equation involving translates $f(ax+b)$. Such equations are common in refinement theory, delay equations, and self-similar probability laws. They generally require boundary, support, or normalization data for uniqueness. The Fabius, up, and signed global forms satisfy related but not identical global equations, so the domain and chosen normalization are part of the statement.

Functional equation. A functional equation constrains a function through values at different arguments, such as reflection, translation, dilation, or convolution identities. The equations $F(1-x) = 1 - F(x)$, $\mathcal{F}'(x) = 2\mathcal{F}(2x)$, and the up-function refinement law are central examples. Functional equations often generate recurrences on special grids and transform products after Fourier or Laplace transformation. Local validity does not automatically imply a global identity.

Fundamental theorem of calculus. The fundamental theorem of calculus links differentiation and integration: a continuous function g has an antiderivative $G(x) = \int_a^x g(t)\ \mathrm{d}t$, and a continuously differentiable F satisfies $F(b) - F(a) = \int_a^b F'(t)\ \mathrm{d}t$. Iteration gives the integral-remainder identity $f(x) = \int_0^x (x-t)^{m-1} f^{(m)}(t)\ \mathrm{d}t/(m-1)!$ when the lower derivatives vanish at zero. Integration by parts, $\int_a^b uv' = [uv]_a^b - \int_a^b u'v$, transfers derivatives between a density and a polynomial; endpoint flatness removes the boundary terms. These forms produce factorial flatness bounds, connect the up-density with the Fabius CDF, and derive moment recurrences.

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7 G

Gamma function. The Gamma function extends the factorial by $\Gamma(s) = \int_0^\infty t^{s-1}\mathrm{e}^{-t}\ \mathrm{d}t$ for $\Re s > 0$ and meromorphic continuation elsewhere. It satisfies $\Gamma(s+1) = s\Gamma(s)$ and has simple poles at the nonpositive integers. Gamma factors arise in Mellin transforms and Gaussian saddle integrals. In the Fabius periodic correction, $\Gamma(-\chi_k)$ combines with a zeta value at a purely imaginary dyadic frequency.

Gamma-zeta coefficient. A Gamma-zeta coefficient is a Fourier coefficient expressed as a product of values of the Gamma and Riemann zeta functions. For the centered Fabius periodic correction, the nonzero coefficient at frequency k has the form $-\Gamma(-\chi_k)\zeta(1-\chi_k)/\log 2$, with $\chi_k = 2\pi \mathrm{i} k/\log 2$. Mellin inversion explains this form: poles repeated along a vertical arithmetic progression become Fourier modes in a logarithmic variable. Exponential Gamma decay makes the periodic fluctuation extremely smooth and numerically small.

Gaussian binomial coefficient. The Gaussian, or q-binomial, coefficient is defined by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}}, \qquad 0 \le k \le n.$$

It is a polynomial in q with nonnegative integer coefficients and specializes to $\binom{n}{k}$ as $q \to 1$. At $q = 1/2$, normalized Gaussian rows act as finite-difference weights on geometric nodes. The Fabius formulas use them to extract leading coefficients, prove translation invariance, and construct complex discrete limits.

Gaussian contraction. For a polynomial $p(X) = \sum_n a_n X^n$, the repository's Gaussian contraction is the linear functional $\mathcal{C}(p) = \sum_n a_n \mu_n$, where $\mu_{2j+1} = 0$ and $\mu_{2j} = (2j-1)!! = (2j)!/(2^j j!)$. Equivalently, $\mathcal{C}(p) = \mathbb{E}[p(Z)]$ for a standard Gaussian $Z \sim N(0,1)$. After a saddle contour is centered and rescaled to its quadratic model, contracting the correction polynomials supplies the mass coefficients of the expansion. This standard-normal auxiliary operation is unrelated to Gaussian q-binomial coefficients.

Gaussian finite difference. A Gaussian finite difference is a q-analogue of an ordinary finite difference, with geometrically spaced nodes and Gaussian binomial weights. Its finite q-binomial theorem implies annihilation of node polynomials below a prescribed degree and an exact response at the leading degree. In the Fabius dyadic evaluation, the base $q = 1/2$ matches the powers of two in the dilation equation, while a second Thue–Morse cancellation acts inside binary blocks.

Gaussian integral. The Gaussian integral is $\int_{-\infty}^{\infty} \mathrm{e}^{-t^2/2}\,\mathrm{d}t = \sqrt{2\pi}$. Its even moments are $\int t^{2m} \mathrm{e}^{-t^2/2}\,\mathrm{d}t = \sqrt{2\pi}(2m-1)!!$, while odd moments vanish. These identities produce the universal coefficients in saddle mass expansions. The reference Gaussian weight is obtained by scaling according to the second derivative of the Fabius phase at the saddle.

Gaussian polynomial. Gaussian polynomial is an older name for a Gaussian, or q-binomial, coefficient, emphasizing that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is a polynomial in q. Do not confuse that usage with the repository's *Gaussian polynomial contraction*, which evaluates a polynomial against standard-normal moments and is used for saddle-expansion polynomials. The q-binomial row-mass and total-variation estimates required by the Toeplitz limit instead live in the discrete-limit modules.

Gaussian rational. A Gaussian rational is a complex number $a + b\mathrm{i}$ with $a, b \in \mathbb{Q}$, that is, an element of $\mathbb{Q}(\mathrm{i})$. It is the fraction field of the Gaussian integers after allowing rational denominators. Exact complex q-binomial approximants can be evaluated in this field when the shift parameter is

Gaussian rational. Scalar-extension lemmas then interpret rational polynomial identities over $\mathbb{C}$ without redoing the arithmetic proof.

Generating function. A generating function encodes a sequence as coefficients of a formal or analytic series. Ordinary generating functions use $\sum a_n z^n$; exponential generating functions use $\sum a_n z^n/n!$. Products encode convolutions, derivatives encode weighted index shifts, and functional equations encode recurrences. Fabius moment sequences, Thue–Morse signs, Bernoulli coefficients, and saddle corrections all have useful generating functions, but their convergence domains and normalizations differ.

Geometric interpolation nodes. The geometric nodes used in the repository are $x_j = -2^j$ for $j = 0, \dots, n$, with constant ratio rather than constant spacing. Lagrange denominators on these nodes factor into q-Pochhammer products, so the leading-coefficient functional is expressed by Gaussian binomial weights with $q = 1/2$. This is the interpolation mechanism behind the finite Fabius q-binomial formulas. A common translation of all nodes changes lower-degree terms but leaves the extracted leading coefficient unchanged.

Germ. A germ at a point is an equivalence class of functions that agree on some neighborhood of that point. Germs let local properties be discussed without fixing a particular neighborhood. A smooth germ records all nearby values, whereas a formal Taylor series records only derivatives at the point; the latter does not determine a smooth germ. The terminating Taylor series of the Fabius function at a dyadic point is therefore compatible with a nonpolynomial, nowhere-analytic germ.

Global binary series. The global binary series is the absolutely convergent representation of the signed Fabius extension obtained by iterating binary Taylor reductions across all scales. Each term depends on a scale coefficient, a reduced local coordinate, and a finite Taylor block. On $[0, 1]$ it equals the bounded Fabius function, while outside that interval it represents the signed continuation. It is global because it is valid for every nonnegative real argument rather than only for dyadic points.

Global differential equation. The global differential equation is $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ for every real x, satisfied by the signed global extension $\mathcal{F}$. The bounded CDF does not satisfy this equation on all of $\mathbb{R}$ because of its constant tails, and the compactly supported up-function satisfies a folded variant. Keeping this distinction explicit prevents invalid differentiation or scaling outside the natural interval. Repeated differentiation of the global equation yields exact derivative scaling.

Global signed extension. The global signed extension $\mathcal{F}$ is the smooth real function extending the local Fabius data so that the pure dilation equation holds on all of $\mathbb{R}$. It can be expressed as a locally finite or absolutely convergent sum of translates weighted by Thue–Morse signs. Unlike the bounded Fabius CDF, it is neither confined to $[0, 1]$ nor monotone. Its integer values and scaling law encode the signs needed for exact dyadic derivatives and global binary reductions.

Grid histogram. A grid histogram is a piecewise-constant or discrete array representation of mass on a regular mesh. Its values may be cumulative counts, normalized bin heights, or samples of a spline. In the Thue–Morse part of the development, a corrected finite grid is identified exactly with a normalized histogram produced by iterated discrete smoothing. This identity bridges combinatorial sign sums and weak or uniform convergence to the continuous Fabius-related density.

Grzegorczyk computability. Grzegorczyk computability for real functions is an early constructive framework based on computable real sequences and effective uniform continuity. A function must transform fast rational names effectively and possess a recursive continuity modulus. This avoids treating arbitrary real inputs as finite data. The repository proves the bounded Fabius function computable in this sense by combining primitive-recursive evaluation of centered finite splines with a uniform approximation theorem and the modulus $d(n) = 2n$.

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8 H

Hadamard finite part. The Hadamard finite part regularizes an integral or divergent expression by subtracting all explicitly known singular terms in its cutoff expansion and retaining the constant term. It handles power and logarithmic divergences and is local to the singular endpoint. Unlike analytic continuation, it is defined directly from asymptotic subtraction, although the two methods often agree when both apply. The Fabius Mellin/Bose analysis uses finite parts at the Mellin origin to isolate the constant and periodic components of a dyadic harmonic sum.

Half moment. A half moment integrates over one half of a symmetric support and therefore retains information discarded by vanishing full odd moments. For the up-function, $d_0 = 1$ and $d_n = n \int_0^1 t^{n-1} \operatorname{up}(t)\, \mathrm{d}t$ for $n \geq 1$. These rational values are related by a finite even-index binomial transform to the full moments. The bridge $F(2^{-n}) = 2^{-\binom{n}{2}} d_n/n!$ converts the integral sequence into exact inverse-dyadic values.

Half-moment numerator. The canonical half-moment numerator G_n is defined by

$$d_n = \frac{G_n}{(n+1)! \prod_{j=1}^{n} (2^j - 1)}.$$

The denominator contains a factorial and odd Mersenne factors; the additional powers of two in inverse-dyadic values enter only through $F(2^{-n}) = 2^{-\binom{n}{2}} d_n/n!$. Clearing the displayed denominator converts the rational recurrence into an integer arithmetic problem whose parity and divisibility results drive exact endpoint valuations.

Half-moment numerator G_n. The integer sequence G_n is the numerator in $d_n = G_n/((n+1)! \prod_{j=1}^{n} (2^j - 1))$. Its triangular recurrence, parity, and divisibility properties are structural inputs to denominator theorems and Reshetnikov integer formulas. It should not be confused with an analytic transform sometimes also denoted $G(z)$. Keeping the full normalization attached to the symbol prevents the half moment, its numerator, and the separately scaled inverse-dyadic sample from being conflated.

Half-moment sequence d_n. The repository's half-moment sequence satisfies $d_0 = 1$ and $d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t$ for $n \geq 1$, together with

$$d_n = \frac{G_n}{(n+1)!\prod_{j=1}^{n}(2^j - 1)}, \qquad F(2^{-n}) = 2^{-\binom{n}{2}}\frac{d_n}{n!}.$$

It is linked to the full moments by a finite even-index binomial transform. For every $n \geq 1$, the reduced numerator and denominator of $2d_n$ are odd, equivalently $v_2(d_n) = -1$; combined with the separate dyadic factor, this yields the exact valuation of $F(2^{-n})$ and the Reshetnikov integrality results.

Half-weighted interval indicator. A half-weighted interval indicator is one in the interior of a closed interval, one half at each endpoint, and zero outside. When adjacent intervals share endpoints, the half weights prevent the pointwise double counting produced by ordinary closed-interval indicators. This convention corrects the endpoints of finite convolution approximants to the up-function and preserves their exact central and partition identities. Endpoints have Lebesgue measure zero, so the convention does not alter integrals, but it does alter pointwise convergence and exact endpoint values.

Hamming weight. The Hamming weight of a binary word or nonnegative integer is the number of one bits. For integers it is the same as the binary digit sum $\mathrm{wt}_2(n)$. Its parity determines the Thue–Morse sign and its exact value counts how many active scales occur in a binary recursion. Algorithms that remove one set bit per step therefore have complexity proportional to Hamming weight rather than to the numerical size of the numerator.

Harmonic number. The n-th harmonic number is $H_n = \sum_{k=1}^{n} 1/k$, with $H_0 = 0$. Harmonic numbers arise in derivatives of Gamma functions, coefficients of logarithms of factorial products, and finite-part expansions. Generalized harmonic numbers use $\sum k^{-r}$. In saddle and Mellin coefficient formulas, they may appear when expanding Gamma ratios or zeta corrections near integer arguments.

HasSum and tsum. In Lean, `HasSum f a` states that the finite partial sums of a family `f` converge to a in the relevant topology. The expression `tsum f` selects that sum when the family is summable and a default value otherwise. Thus a `HasSum` theorem retains the convergence witness, while an equality involving `tsum` is a convenient value statement whose meaning depends on summability. The repository deliberately distinguishes these forms for Fourier, translate, Legendre, and binary-reduction series so that an equality does not hide the existence or mode of convergence of the infinite sum.

Highest-set-bit recursion. A highest-set-bit recursion decomposes a positive integer a as $2^m + r$ with $0 \leq r < 2^m$, processes the leading binary block, and recurses on a smaller integer. For Fabius dyadic evaluation, reflection and Taylor reduction turn this decomposition into an exact arithmetic algorithm. Each recursive call removes at least one set bit, providing a transparent termination measure and an operation count controlled by the binary representation.

Hockey-stick identity. The hockey-stick identity is $\sum_{j=r}^{n}\binom{j}{r} = \binom{n+1}{r+1}$. Its name comes from the shape of the selected entries in Pascal's triangle. Repeated prefix summation and discrete B-spline formulas often reduce to this identity, as do sums over ordered compositions with a final free part. The repository uses it to collapse nested finite sums arising from Thue–Morse cancellation and spline integration.

Holomorphic function. A holomorphic function is complex differentiable at every point of an open domain. Complex differentiability implies analytic power-series expansion and strong rigidity properties such as the identity theorem and Cauchy's integral formula. Entire functions are holomorphic on all of $\mathbb{C}$. The complex moment-generating function and Fourier transform of the up-function are holomorphic objects; contour and Mellin arguments rely on their domains, singularities, and growth estimates.

Homothety. A homothety is a dilation about a fixed center, $x \mapsto c+a(x-c)$. Homothetic copies of a compactly supported function are rescaled and shifted versions that preserve shape while changing support. Rvachev's atomic-function theory builds approximation systems from homothetic copies of the up-function. The Fabius dilation equation is centered at zero and uses ratio two, while folding and translating connect it to support-centered copies.

Horner scheme. Horner's scheme evaluates a polynomial $a_0 + a_1x + \cdots + a_nx^n$ as $a_0 + x(a_1 + x(\cdots + xa_n))$. It uses n multiplications and n additions and reduces intermediate expression growth. Exact dyadic formulas involving moment polynomials can be implemented in Horner form to avoid repeated exponentiation. In rational arithmetic it also helps control numerator and denominator sizes, although bit complexity still depends on coefficient growth.

Hypergeometric term. A sequence term a_n is hypergeometric when the ratio a_{n+1}/a_n is a rational function of n. Factorials, binomial coefficients, and double factorials are basic examples. Their sums can often be manipulated by telescoping or symbolic recurrence methods. The Fabius arithmetic formulas combine hypergeometric prefactors with recursively defined moment numerators; the q-analogues replace rational dependence on n by rational dependence on q^n.

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9 I

Identity theorem. The identity theorem says that two holomorphic functions on a connected domain that agree on a set with an accumulation point agree everywhere. A real-analytic analogue holds on intervals. It is used conceptually in nowhere-analytic arguments: if a smooth Fabius germ were analytic and agreed with a terminating Taylor polynomial on a neighborhood, overlap and continuation would force polynomial behavior incompatible with the global scaling and shape.

Improper integral. An improper integral is defined as a limit when the interval is unbounded or the integrand is singular at an endpoint. Convergence requires the cutoff integrals to approach a finite limit. Absolute convergence is stronger and permits more transformations. Laplace, Mellin, and Fourier integrals in the Fabius theory are often improper; when a singular Mellin term prevents ordinary convergence, a finite-part integral rather than an improper integral is used.

Independent random variables. Random variables are independent when probabilities of joint events factor, equivalently when expectations of suitable products factor. Independence implies that characteristic and moment-generating functions multiply under addition and that probability laws convolve. The Fabius distribution arises from an infinite sum of independent uniform variables at dyadically decreasing scales. This representation explains the infinite product transform and produces finite convolution splines by truncation.

Indicator function. The indicator of a set A is $\mathbf{1}_A(x) = 1$ for $x \in A$ and 0 otherwise. Normalized interval indicators are uniform probability densities. Their repeated convolutions are splines, and their Fourier transforms are phase factors times sinc functions. The finite approximants to the up-function begin with scaled interval indicators. For pointwise endpoint identities the development uses a half-weighted closed-interval version; although the two choices agree almost everywhere, their boundary values are not interchangeable.

Infinite convolution. An infinite convolution is a limit of finite convolutions of probability measures or functions. For probability laws it represents the distribution of an infinite sum of independent random variables, provided the partial sums converge. Weak convergence is often the natural initial topology, while additional smoothness and tail control may yield densities and uniform convergence. The Fabius law is the infinite convolution of dyadically scaled uniform laws; its Fourier transform is the corresponding infinite product of sinc factors.

Infinite product. An infinite product $\prod_{n\geq 0}(1 + u_n)$ converges in the nonzero sense when its partial products tend to a nonzero limit. The sufficient condition $\sum_n |u_n| < \infty$ applies when no factor vanishes; products tending to zero require a separately stated convention. Locally uniform convergence of holomorphic factors produces a holomorphic limit, with zeros inherited from the finite factors when accumulation is controlled. The up-function transform is an entire sinc product whose zero structure, logarithm, and negative-real specialization drive the Fourier and endpoint theories.

Infinite random series. An infinite random series is a sum $\sum_{n\geq 0} X_n$ of random variables defined as a limit of partial sums in a chosen mode, such as almost surely or in L^2. If X_n are uniformly bounded by an absolutely summable deterministic sequence, convergence is immediate for every outcome. The Fabius variable is a weighted series of independent uniforms with geometric weights, so it has compact support and an entire moment-generating function.

Infinite weighted sum. An infinite weighted sum has the form $\sum a_n x_n$, with convergence governed by the weights and the size of the terms. In probability, independent bounded variables with $\sum |a_n| < \infty$ give almost-sure absolute convergence. Dyadic weights make the tail comparable to the next scale and create exact self-similarity. This is the probabilistic source of the Fabius dilation equation and of the finite spline truncations.

Inflection point. An inflection point is a point where the local convexity changes from convex to concave or conversely. A zero of the second derivative alone is not sufficient unless a sign change or equivalent convexity statement is proved. The bounded Fabius function has a unique inflection point at $x = 1/2$: its derivative is strictly increasing on the left half and strictly decreasing on the right half. Symmetry forces the location, while strict unimodality of the up-density gives uniqueness.

Injective, surjective, and bijective. A map is injective if equal outputs imply equal inputs, surjective if every target value is attained, and bijective if both hold. Strict monotonicity gives injectivity on an interval; continuity and endpoint values often give surjectivity onto the interval between them. The Fabius restriction $F : [0, 1] \to [0, 1]$ is bijective, so it has a well-defined inverse. Globally, the constant tails destroy injectivity, which is why inverse statements always specify the restricted domain.

Integer composition. A composition of a positive integer n is an ordered tuple of positive integers $(m_1, \ldots, m_r)$ with sum n. Unlike a partition, order matters. Weak compositions allow zero parts. Compositions index coefficients of powers and logarithms of formal series: the coefficient of z^n in $U(z)^r$ is a sum over r-part compositions of n. The all-orders Fabius saddle coefficients use such sums explicitly.

Integer partition. An integer partition of n is a multiset of positive integers whose sum is n, with order ignored. Partitions classify multiplicity patterns in Bell-polynomial and Faà di Bruno expansions. They are distinct from partitions of unity, which are families of functions summing to one, and from ordered compositions. In all-orders coefficient formulas one may pass between partition-indexed multiplicities and composition-indexed ordered products depending on which is easier to formalize.

Integrability. Integrability means that a function has a well-defined integral in the relevant sense, usually Lebesgue or Riemann. Absolute integrability, $\int |f| < \infty$, ensures stable Fourier transforms and interchange of sums and integrals. Compactly supported continuous functions are absolutely integrable, as are the up-function and its derivatives. Mellin kernels may fail to be integrable at zero until singular terms are subtracted.

Integral transform. An integral transform maps f to a new function $Tf(s) = \int K(s, x) f(x)\ \mathrm{d}x$ using a kernel K. Fourier, Laplace, Mellin, and moment transforms emphasize different structures: translation, exponential decay, scaling, and polynomial moments. The Fabius theory moves among all four. Each transform has its own convergence region and inversion theorem, so identities must be transferred with explicit hypotheses.

Integrality. An integrality theorem proves that a rationally defined expression is in fact an integer. Such results often require divisibility, valuation, or recurrence arguments that are invisible in the original quotient. The repository proves positivity and oddness of normalized Reshetnikov integers and constructs division-free numerator recurrences. Integrality is stronger than rationality and enables modular or parity conclusions that feed back into exact denominator formulas.

Interior of the support. The interior of the support is the largest open set contained in the support. For the up-function, the support is $[-1,1]$ and its interior is $(-1,1)$, where the function is strictly positive. Boundary points belong to the topological support even though the value and every derivative vanish there. Distinguishing support from its interior is important in statements about positivity, analyticity, and zero sets.

Interpolating polynomial. Given distinct nodes $x_0,\dots,x_n$ and values y_j, the interpolating polynomial is the unique polynomial of degree at most n satisfying $p(x_j)=y_j$. Lagrange's formula writes it as a sum of basis polynomials. On geometric nodes, its leading coefficient is a Gaussian-weighted functional of the data. The Fabius q-binomial identities identify this leading coefficient after Thue–Morse cancellation with an exact dyadic value.

Inverse-dyadic value. In the repository, an inverse-dyadic value means a sample at an inverse power of two, $F(2^{-n})$, not a value of the inverse function. The exact bridge to the half-moment sequence is

$$F(2^{-n}) = 2^{-\binom{n}{2}}\frac{d_n}{n!}, \qquad n \geq 0.$$

For $n \geq 1$, its exact dyadic valuation is $v_2(F(2^{-n})) = -\binom{n}{2} - v_2(n!) - 1$. These samples seed composition/path recurrences, exact denominator formulas, and endpoint asymptotics. A preimage $F^{-1}(a/2^m)$ should instead be called an inverse-function value at a dyadic ordinate.

Inverse function. If $f : I \to J$ is bijective, its inverse $f^{-1} : J \to I$ satisfies $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. For a continuous strictly monotone map between intervals, the inverse is continuous; if $f' \neq 0$, it is differentiable locally with derivative $1/f'$. The Fabius inverse is defined on $[0,1]$ using the strictly increasing bounded function. Near the endpoints its behavior is governed by inverting the corrected flat asymptotic.

Inverse-Lambert expansion. An inverse-Lambert expansion rewrites the solution of an equation such as $\lambda 2^{-\lambda} = x$ in elementary logarithmic terms. Exactly, $\lambda = -W_{-1}(-x\log 2)/\log 2$. Iterating the relation $\lambda \log 2 = |\log x| + \log\lambda$ gives nested logarithms and a full asymptotic series. The repository formalizes the coefficient algebra and remainder bounds needed to transfer the sharp saddle expansion from λ to powers of $1/|\log x|$.

Inverse-logarithmic error. An inverse-logarithmic error is a remainder bounded by a negative power of a growing logarithm, for example $\mathcal{O}(1/|\log x|)$ as $x \downarrow 0$. Such errors decay very slowly and are therefore sensitive to bounded oscillatory terms: a missing nonconstant periodic contribution is not swallowed by them. The corrected Fabius asymptotic attains an inverse-logarithmic error only after evaluating the periodic fluctuation at the exact lower-Lambert phase.

Inverse-power table. The inverse-power table through level n is the finite collection $(F(1), F(2^{-1}), \dots, F(2^{-n}))$. Each entry is rational and is obtained exactly from half moments or a terminating recurrence; collectively the table supplies base values for fast binary evaluation and regression tests. Its entries also have normalized odd-integer forms and sharp asymptotics. The term “inverse power” refers to the arguments 2^{-j}, not to reciprocal function values or to the inverse Fabius function.

Inversion formula. An inversion formula reconstructs an original object from a transform or cumulative operation. Fourier, Laplace, and Mellin transforms each have inversion integrals; finite differences invert prefix summation; binomial inversion undoes a binomial transform. In the Fabius directory, inversion formulas recover the up-density from its sinc product and the endpoint function from a Bromwich contour. Each inversion requires its own convergence and regularity hypotheses.

IsFabius characterization. `IsFabius` is the Lean predicate collecting the axioms that characterize the bounded Fabius function: constant tails, reflection symmetry, differentiability or smoothness, and the left-half differential equation in the exact repository formulation. Mathematically, it defines a property of candidate functions rather than a new function. Existence supplies one witness and uniqueness proves that any two witnesses agree, allowing later theorems to be stated abstractly for every function satisfying the characterization.

Iterated derivative. The iterated derivative $f^{(m)}$ is obtained by differentiating m times. For a self-similar differential equation, each iteration introduces both a coefficient and a chain-rule scaling. The signed Fabius extension satisfies the closed formula $\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x)$. This turns an apparently analytic hierarchy into exact arithmetic at dyadic points and yields the sharp derivative suprema.

Iterated prefix sum. For a sequence a_n, its prefix sum is $A_n = \sum_{k\leq n} a_k$; iterating this operation produces higher discrete antiderivatives. Repeated prefix sums of an indicator sequence yield binomial coefficients and discrete B-splines. When applied to Thue–Morse signs, low-order sums vanish over complete dyadic blocks by Prouhet cancellation, and the first nonzero iterated sum has a structured polynomial form. This connects binary combinatorics to spline limits.

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10 J

Jet. The m-jet of a smooth function at a point is the finite list of derivatives through order m, often encoded as its Taylor polynomial modulo terms of degree $m+1$. Jets capture the local data needed for finite-order approximation without asserting convergence of the full Taylor series. Saddle expansions use phase and amplitude jets at a moving critical point; formal recurrences combine these jets into all-orders coefficients. At a dyadic Fabius point the infinite jet is eventually zero, yet the function is not locally polynomial.

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11 K

K-fold convolution. A K-fold convolution combines K functions or measures by repeated convolution. For probability laws it is the distribution of a sum of K independent variables. In general its support is contained in the closure of the Minkowski sum of the individual supports; equality holds for the compactly supported positive interval laws used in the Fabius construction, but signed or complex cancellation can make the inclusion strict. K-fold identities involving up-functions and Thue–Morse coefficients generalize the basic self-convolution relation, while transforms reduce them to products or powers.

K-fold Thue-Morse sum. A K-fold Thue–Morse sum is a multiple discrete convolution or signed sum built from the sequence $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}$. Its generating function is a power of the finite Thue–Morse product, so high-order zeros at one translate into vanishing moments and Prouhet-type cancellation. In the repository such sums are related to convolution powers, discrete splines, and scaled limits of the Fabius/up functions.

Kernel. A kernel is a function that defines an integral, convolution, summation, or transformation operator. Examples include the Fourier exponential, the Laplace exponential, interval indicators, and Bose-Einstein kernels. Kernel properties such as positivity, normalization, support, symmetry, and decay determine the operator's behavior. The Fabius development uses probability kernels for the fixed point, sinc factors as transform kernels, and regularized Mellin kernels for endpoint asymptotics.

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12 L

L2 norm. For a square-integrable function on a measure space, the L^2 norm is $\|f\|_2 = (\int |f|^2)^{1/2}$. It comes from the inner product $\langle f, g\rangle = \int f\overline{g}$, so orthogonality and Pythagorean identities are available. Fourier-Legendre partial sums are orthogonal projections in $L^2[-1,1]$ and therefore give the unique least-squares polynomial approximants. Uniform convergence proved later is stronger than convergence in this norm.

Lagrange interpolation. Lagrange interpolation writes the unique degree-at-most-n polynomial through distinct data (x_j, y_j) as $p(x) = \sum_{j=0}^{n} y_j \ell_j(x)$, where $\ell_j(x) = \prod_{m\neq j}(x - x_m)/(x_j - x_m)$. Its leading coefficient is $\sum_j y_j / \prod_{m\neq j}(x_j - x_m)$. For geometric nodes -2^j, these denominators simplify to q-Pochhammer products and Gaussian binomial weights, revealing the interpolation structure of exact Fabius dyadic formulas.

Lambert W function. The Lambert W function is the multivalued inverse of $w \mapsto w\mathrm{e}^w$: it satisfies $W(z)\mathrm{e}^{W(z)} = z$. On $(-\mathrm{e}^{-1}, 0)$ there are two real branches, $W_0 \in (-1, 0)$ and $W_{-1} \in (-\infty, -1)$. The Fabius endpoint reference scale diverges as $x \downarrow 0$, so the lower branch W_{-1}, not the principal branch, gives the relevant coordinate. Its nested-log expansion translates the exact lower-Lambert coordinate into elementary logarithms.

Laplace transform. The Laplace transform of a function or measure on a half-line is $\mathcal{L}f(s) = \int_0^\infty \mathrm{e}^{-sx} f(x)\,\mathrm{d}x$, on the region where the integral converges. For a compactly supported probability law it extends to an entire moment-generating function after a sign change. Convolution becomes multiplication. The Fabius transform factors into elementary terms from the independent uniform summands, and its large negative-real behavior is the starting point for the corrected endpoint asymptotic.

Laurent expansion. A Laurent expansion represents a holomorphic function on an annulus as $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$. Finitely many negative powers characterize a pole; the coefficient of $(z-z_0)^{-1}$ is the residue. Mellin transforms in the Fabius analysis have Laurent expansions at singular points such as the origin or $s=1$. The finite part is the constant coefficient after the principal polar terms are removed.

Leading-coefficient functional. On polynomials of degree at most n, the leading-coefficient functional sends $p(x) = a_n x^n + \cdots$ to a_n. At distinct nodes $x_0, \ldots, x_n$, Lagrange interpolation gives

$$[x^n]p = \sum_{j=0}^{n} \frac{p(x_j)}{\prod_{m\neq j}(x_j - x_m)}.$$

For geometric nodes the weights are normalized Gaussian binomial coefficients. Translating every node by the same scalar changes only lower coefficients, so the resulting numerator polynomial in the common shift is constant; this explains the repository's rational, real, complex, and Gaussian-rational shift independence.

Least common multiple. The least common multiple $\operatorname{lcm}(a_1, \ldots, a_m)$ is the smallest positive integer divisible by each a_j. Prime valuations satisfy $v_p(\operatorname{lcm}(a_j)) = \max_j v_p(a_j)$. LCM products give economical common denominators for families of rational moment or dyadic values. In the Fabius arithmetic layer, they control odd prime factors separately from the exactly known power of two.

Least Lipschitz constant. The least Lipschitz constant of f is $\sup_{x\neq y} |f(x) - f(y)|/|x-y|$, possibly infinite. For a continuously differentiable function on an interval it equals $\sup |f'|$. The bounded Fabius function is globally Lipschitz with optimal constant 2: the derivative bound gives the upper estimate, and equality is attained at the midpoint because $F'(1/2) = 2$. Optimality matters for the sharp effective modulus of continuity.

Least-squares approximation. A least-squares approximation minimizes the squared L^2 error $\int |f-p|^2$ over a finite-dimensional approximation space. In an orthogonal basis, the minimizer is obtained by truncating the corresponding Fourier coefficients. The Legendre partial sum of degree n is therefore the unique best polynomial approximation to the up-function in $L^2[-1,1]$. This optimality is metric-specific and does not by itself imply best uniform approximation.

Lebesgue integral. The Lebesgue integral integrates measurable functions by measuring level sets and is stable under limit theorems such as monotone and dominated convergence. It naturally handles probability measures and almost-everywhere equivalence. Moments, expectations, convolution measures, and transform integrals in the Fabius development are most naturally Lebesgue integrals, even when continuity and compact support make them agree with classical Riemann integrals.

Legendre coefficient. A Legendre coefficient is the component of a function along the Legendre polynomial P_n under the inner product on $[-1, 1]$. With $\int_{-1}^{1} P_n^2 = 2/(2n+1)$, the coefficient is $(2n+1)\int f P_n/2$. Symmetry annihilates odd coefficients for an even function. The Fabius directory expresses the up-function coefficients exactly through moments and proves bounds strong enough for absolute uniform convergence.

Legendre polynomial. The Legendre polynomial P_n is the degree-n polynomial normalized by $P_n(1) = 1$ and orthogonal to lower-degree polynomials on $[-1, 1]$ with unit weight. Rodrigues' formula is $P_n(x) = \frac{1}{2^n n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2 - 1)^n$, and it solves $((1 - x^2)P_n')' + n(n+1)P_n = 0$. These properties support integration by parts, coefficient recurrences, and least-squares approximation of the up-function.

Legendre's factorial-valuation formula. For a prime p, Legendre's formula states

$$v_p(n!) = \sum_{j\geq 1}\left\lfloor \frac{n}{p^j}\right\rfloor = \frac{n - s_p(n)}{p-1},$$

where $s_p(n)$ is the base-p digit sum; only finitely many terms in the sum are nonzero. At $p = 2$, this becomes $v_2(n!) = n - \mathrm{wt}_2(n)$, directly linking factorial divisibility to binary weight. The inverse-dyadic valuation theorem uses this identity to rewrite the exact power of two in $F(2^{-n})$. This result is unrelated to Legendre polynomials despite the shared name.

Leibniz rule. The Leibniz rule for the n-th derivative of a product is $(fg)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(k)} g^{(n-k)}$. It is the differential analogue of binomial convolution. Formalized moment, transform, and saddle calculations repeatedly use it to expand products with exact finite index ranges. For exponential products it combines naturally with Bell polynomials and cumulants.

Levy continuity theorem. Levy's continuity theorem states that pointwise convergence of characteristic functions to a function continuous at zero is equivalent to weak convergence of the corresponding probability measures. It is a standard route from finite convolution products to an infinite probability law. In the Fabius construction, the explicit convergence of the sinc products can identify the weak limit and its characteristic function.

Lipschitz function. A function is Lipschitz with constant L when $|f(x) - f(y)| \leq L|x - y|$ for all relevant x, y. Lipschitz continuity is stronger than uniform continuity and implies absolute continuity on intervals. A bounded derivative gives a Lipschitz bound by the mean value theorem. The Fabius function has least global constant 2, while related spline approximants inherit uniform Lipschitz control.

Local phase. The local phase is the Taylor expansion of the exponent around the chosen central point of a contour integral. At an exact saddle its linear term vanishes, leaving a negative quadratic Gaussian term followed by cubic and higher corrections. At the repository's lower-Lambert reference radius the full linear term need not vanish; it is retained with the cubic and other odd corrections and bounded quantitatively. The Fabius modules isolate this local phase from the global periodic factor, prove a central approximation, and control the remainder on a uniformly valid window.

Locally finite sum. A sum of functions $\sum_j f_j(x)$ is locally finite if every point has a neighborhood meeting the support of only finitely many summands. Such a sum is unambiguously defined and inherits continuity or smoothness termwise without convergence estimates. Translate sums of compactly supported up-functions are often locally finite. The signed global Fabius extension also has presentations where only finitely many translated support pieces contribute at a fixed point, alongside other genuinely infinite absolutely convergent series.

Locally uniform convergence. A sequence converges locally uniformly on a domain if it converges uniformly on every compact subset. For holomorphic functions, locally uniform convergence preserves holomorphy and permits termwise differentiation. Infinite products are also commonly controlled locally uniformly. The sinc product for the up-function's Fourier transform and the negative-Laplace logarithmic series are established on compact subsets of their natural domains before global growth or tail estimates are considered.

Log-periodic function. A function of $x > 0$ is log-periodic with ratio $b > 1$ if it is periodic in $\log_b x$, equivalently $Q(bx) = Q(x)$. Such oscillations arise in discrete-scale-invariant problems because scaling by b advances the logarithmic phase by one. The Fabius endpoint correction is one-periodic in a lower-Lambert coordinate rather than exactly in $\log_2 x$, but it has the same dyadic origin. A log-periodic term can be tiny yet prevent convergence to a constant asymptotic ratio.

Log-squared asymptotic. A log-squared asymptotic has a dominant term proportional to $(\log(1/x))^2$, often in the logarithm of a very flat function. For the Fabius endpoint, $\log F(x)$ has a leading negative quadratic logarithmic scale, modified by lower nested-log, constant, and periodic contributions. This explains decay faster than every power of x but slower than typical $\exp(-c/x^\alpha)$ scales. The exact coefficient is determined by the dyadic dilation factor $\log 2$.

Logarithmic delay equation. A logarithmic change of variables converts multiplicative scaling into additive translation. If $x = \mathrm{e}^{-t}$, then $f(2x)$ becomes a shifted function of $t - \log 2$. The resulting relation is a delay or advance equation with fixed step $\log 2$. This viewpoint explains why periodic functions of a logarithmic coordinate appear as homogeneous or residual solutions in the Fabius endpoint analysis.

Logarithmic scale. A logarithmic scale uses $L = \log(1/x)$, $\log L$, and inverse powers of these quantities to organize a limit as $x \downarrow 0$. It is appropriate when the decisive saddle grows only logarithmically. The Fabius expansion is most exact in $\lambda = -W_{-1}(-x \log 2)/\log 2$, but λ itself has a nested-log expansion. Distinguishing the exact coordinate from its elementary logarithmic approximation prevents phase errors.

Lower Lambert branch. The lower real branch W_{-1} of Lambert W is defined on $[-\mathrm{e}^{-1}, 0)$, takes values in $(-\infty, -1]$, and satisfies $W_{-1}(z)\mathrm{e}^{W_{-1}(z)} = z$. As $z \uparrow 0$, it tends to $-\infty$. This branch solves the large positive reference-balance equation used to organize the Fabius endpoint saddle regime. Using the principal branch would produce a bounded coordinate and the wrong asymptotic scale.

Lower Lambert coordinate. For small $x > 0$, the repository defines the lower-Lambert phase by

$$\lambda(x) = -\frac{W_{-1}(-x \log 2)}{\log 2}, \qquad \lambda(x) 2^{-\lambda(x)} = x.$$

It is an explicit reference saddle coordinate, tends to infinity as $x \downarrow 0$, and advances by approximately one under dyadic rescaling. The periodic correction and all inverse-power coefficients are most naturally evaluated at $\lambda(x)$, not at a truncated logarithmic surrogate. This terminology packages the useful coordinate without asserting that λ itself is the contour's exact critical point.

Lower-order term. A lower-order term is asymptotically smaller than the leading term in a specified scale. The phrase is meaningful only after the comparison variable and mode of comparison are fixed. A bounded periodic term is lower order than a quadratic logarithm but is not lower order than an $o(1)$ remainder in the exponent. The Fabius correction illustrates why informal claims that a term is negligible must be matched to the claimed error scale.

Lucas's theorem. If p is prime and $n = \sum_j n_j p^j$, $k = \sum_j k_j p^j$, then Lucas's congruence is

$$\binom{n}{k} \equiv \prod_j \binom{n_j}{k_j} \pmod{p}.$$

For $p = 2$, $\binom{n}{k}$ is odd exactly when every set bit of k is also a set bit of n. Consequently row n of Pascal's triangle has $2^{\mathrm{wt}_2(n)}$ odd entries. The Fabius arithmetic uses this parity count to prove oddness of normalized moment numerators and hence exact 2-adic valuations.

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13 M

Maclaurin series. A Maclaurin series is a Taylor series centered at zero. When the function is analytic there,

$$f(z) = \sum_{n \geq 0} \frac{f^{(n)}(0)}{n!} z^n.$$

Moment-generating and Fourier transforms of compactly supported laws have entire Maclaurin series whose coefficients are moments. By contrast, the bounded Fabius function itself has the zero Taylor series at the flat endpoint 0 but is positive immediately to the right, a standard demonstration that smoothness does not imply analyticity.

Major arc. In a contour or oscillatory integral, a major arc is the neighborhood of the dominant saddle or stationary point where the main contribution is extracted. Its width is chosen to make a local Taylor expansion accurate while retaining nearly all Gaussian mass. The complement is the minor arc or tail. Fabius saddle modules prove separate central estimates on the major arc and exponential or comparison bounds outside it, then balance the chosen radius to obtain a uniform remainder.

Mass expansion. A mass expansion is the asymptotic series for the normalized integral remaining after the leading exponential value and Gaussian scale have been factored out. It typically begins $1 + a_1/\lambda + a_2/\lambda^2 + \cdots$. The coefficients combine phase and amplitude jets and Gaussian moments. Taking the formal logarithm of this series gives the additive coefficients in $\log F(x)$. The repository provides both recurrences and explicit ordered-composition formulas for these saddle mass coefficients.

Mean value theorem. The mean value theorem states that a continuous function on $[a, b]$, differentiable on (a, b), satisfies $f(b) - f(a) = f'(c)(b - a)$ for some c. Consequently $|f(b) - f(a)| \leq \sup |f'| \, |b - a|$. Iterating such estimates with the exact derivative scaling yields one form of effective Fabius flatness. It also proves Lipschitz continuity and turns derivative maxima into sharp global constants.

Measure-zero set. A set has Lebesgue measure zero if it can be covered by intervals of arbitrarily small total length. Statements holding outside such a set hold almost everywhere. Countable sets, including the dyadic rationals, have measure zero even when dense. In analytic-locus discussions this distinction matters: exceptional behavior on every dyadic point may be topologically dense yet measure-theoretically negligible. The Fabius nowhere-analytic result is stronger than an almost-everywhere statement.

Mellin finite part. A Mellin finite part is the constant term obtained when a Mellin transform has a pole at the point of interest. One subtracts the polar Laurent terms or equivalently regularizes the original endpoint integral. In the Fabius negative-Laplace decomposition, the Mellin origin carries the polynomial-logarithmic main terms, while shifted imaginary poles generate the periodic correction. Correct extraction of the finite part fixes the endpoint constant and the centering of the periodic function.

Mellin inversion. Mellin inversion reconstructs a function from $M(s) = \int_0^\infty f(x) x^{s-1} \, \mathrm{d}x$ by $f(x) = (2\pi \mathrm{i})^{-1} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} M(s) x^{-s} \, \mathrm{d}s$, under suitable hypotheses. Shifting the contour across poles produces asymptotic expansions. Periodically spaced imaginary poles yield Fourier series in $\log x$. This is the conceptual source of the Gamma–zeta periodic correction in the Fabius Laplace logarithm.

Mellin transform. The Mellin transform is $M_f(s) = \int_0^\infty f(x) x^{s-1} \, \mathrm{d}x$. It converts multiplicative scaling into multiplication by powers and is the Fourier transform in logarithmic coordinates. Harmonic sums over dyadic scales become products with $(1 - 2^{-s})^{-1}$, whose imaginary lattice of poles creates one-periodic functions. The Fabius endpoint analysis applies Mellin methods to a regularized Bose-type kernel derived from the negative-Laplace product.

Meromorphic continuation. Meromorphic continuation extends a function beyond its initial domain while allowing isolated poles. Gamma and zeta functions are standard examples. Mellin transforms often converge only in a vertical strip but continue meromorphically through functional identities or subtraction of endpoint asymptotics. In the Fabius analysis, locating the poles and computing their residues is more important than merely knowing continuation exists, because the pole at zero and the imaginary dyadic lattice determine the main and periodic terms.

Mersenne factor. A Mersenne factor is an integer of the form $2^m - 1$. Products of odd Mersenne factors arise when the half-base q-Pochhammer symbol $(1/2;1/2)_n = \prod_{j=1}^{n}(1-2^{-j})$ is cleared of powers of two. These products appear in moment denominators, Gaussian binomial normalizations, and Reshetnikov integers. Their oddness makes them especially convenient for separating 2-adic valuations from odd divisibility.

Minkowski sum of supports. For subsets $A, B \subset \mathbb{R}$, their Minkowski sum is $A + B = \{a + b : a \in A, b \in B\}$. In general $\operatorname{supp}(f * g) \subseteq \overline{\operatorname{supp} f + \operatorname{supp} g}$; if the supports are compact the sum is already closed. Equality holds for the positive, nondegenerate interval laws in the Fabius construction, whereas signed or complex cancellation can make the inclusion strict. Consequently finite sums of scaled uniforms have support equal to the sum of their interval supports, fixing the centering and endpoints of the spline approximants.

Minor arc. The minor arc is the portion of an oscillatory or contour integral outside the central saddle neighborhood. A successful saddle-point proof shows that its contribution is smaller than the target remainder, often by a strict loss in the real part of the phase or by transform decay. The Fabius development treats central, tail, and reference-weight estimates in separate modules, preventing a local Gaussian expansion from being mistaken for a complete contour proof.

Modulus of continuity. A modulus of continuity is a nondecreasing function $\omega(\delta) \to 0$ such that $|f(x) - f(y)| \le \omega(|x - y|)$. A Lipschitz function has $\omega(\delta) = L\delta$. An effective modulus must be computable in a specified representation. For the Fabius function, the exact least Lipschitz constant gives both a sharp analytic modulus and a simple recursive modulus used in computability.

Moment. The n-th moment of a random variable or finite measure is $\mathbb{E}[X^n]$ or $\int x^n \, \mu(\mathrm{d}x)$, when finite. Moments encode location, scale, and shape; for compactly supported laws all moments exist and determine the law. The Fabius development uses full moments of the centered up-density and half moments on one side of the support. Their triangular recurrences and generating functions control exact dyadic values and Legendre coefficients.

Moment-generating function. The moment-generating function is $M_X(z) = \mathbb{E}(\mathrm{e}^{zX})$; for bounded X it is entire and its Taylor coefficients are moments. For the Fabius variable $X = \sum_{j\ge1} 2^{-j}U_j$,

$$M_X(z) = \prod_{j\ge1} \frac{\mathrm{e}^{z/2^j} - 1}{z/2^j}.$$

Its imaginary restriction includes the centering phase: $M_X(\mathrm{i}t) = \mathrm{e}^{\mathrm{i}t/2} \prod_{j\ge1} \operatorname{sinc}(t/2^{j+1})$. The pure product $\prod_{j\ge1} \operatorname{sinc}(t/2^j)$ is instead the characteristic function of $2X - 1$. Independence explains the products, while the negative-real specialization drives endpoint analysis.

Moment numerator. The repository uses distinct integer numerators F_n and G_n for its rational full and half moments:

$$c_n = \frac{F_n}{(2n+1)!!\prod_{j=1}^{n}(2^{2j}-1)}, \qquad d_n = \frac{G_n}{(n+1)!\prod_{j=1}^{n}(2^{j}-1)}.$$

These displayed denominators contain factorial or double-factorial terms and odd Mersenne factors; powers of two enter the separate bridge to $F(2^{-n})$. Integer recurrences for F_n, G_n expose parity, positivity, and divisibility hidden in the raw rational moments.

Moment numerator F_n. The arithmetic sequence F_n is defined by $c_n = F_n/((2n+1)!!\prod_{j=1}^{n}(2^{2j}-1))$. It is distinct from the function $F(x)$, despite sharing the letter in source conventions. Its triangular recurrence and divisibility properties organize exact full-moment arithmetic. The glossary uses the phrase "moment numerator F_n" whenever ambiguity with the bounded Fabius function is possible.

Moment recurrence. A moment recurrence expresses a moment of order n in terms of lower moments, often by expanding a distributional fixed-point equation. If X has the same law as a scaled combination of X and an independent elementary variable, applying $\mathbb{E}[(\cdot)^n]$ and the binomial theorem yields a triangular relation. The Fabius full and half moment recurrences are rational and exact, and after normalization become division-free integer recurrences.

Moment sequence. A moment sequence is a sequence $m_n = \int x^n \,\mu(\mathrm{d}x)$ associated with a measure. Not every sequence is a moment sequence; positivity constraints and growth conditions are involved. Compact support makes the moment problem determinate. The repository's sequence c_n records even centered moments, while d_n records normalized half moments. Their arithmetic normalizations are used throughout the dyadic and Legendre layers.

Monotone convergence theorem. The monotone convergence theorem states that if $0 \le f_n \uparrow f$ pointwise, then $\int f_n \uparrow \int f$, allowing the value $+\infty$. It is useful for increasing approximations by positive densities or distribution functions. Fabius finite probability constructions often allow stronger dominated or uniform convergence, but monotone convergence remains a foundational tool for nonnegative integral limits.

Monotone function. A function is nondecreasing when $x < y$ implies $f(x) \le f(y)$, and strictly increasing when the inequality is strict. Every cumulative distribution function is nondecreasing. The bounded Fabius function is strictly increasing on $[0, 1]$, because its density is positive in the interior, but it is constant on the exterior tails. The up-function is not globally monotone; it increases to the origin and then decreases.

Multiplicative periodic fluctuation. A multiplicative periodic fluctuation is a bounded factor whose value depends periodically on a logarithmic phase, typically $\exp(\Psi(\lambda(x)))$. In logarithmic asymptotics it appears additively as Ψ. Because it need not approach a limit, it can prevent a ratio from converging even when all coarse logarithmic orders agree. The corrected Fabius endpoint formula contains exactly such a factor, with very small but explicitly nonzero Fourier modes.

14 N

Negative-Laplace decomposition. The negative-Laplace decomposition analyzes $\log M(-s)$ for large $s > 0$, where M is the Fabius moment-generating function. An exact dyadic product is converted into a harmonic sum of the kernel $\kappa(u) = \log((1-\mathrm{e}^{-u})/u)$. Mellin analysis then separates explicit polynomial-logarithmic terms, a constant, a smooth one-periodic correction, and a decaying remainder. This decomposition is the global input for the Bromwich saddle method.

Negative-Laplace product. The negative-Laplace product is the exact factorization of the Fabius transform on the negative real axis, for example $G(-s) = \prod_{j\geq 1}(1-\mathrm{e}^{-s/2^j})/(s/2^j)$. Each factor is the Laplace transform of a uniform summand after normalization. The product is positive for $s > 0$, so its real logarithm is unambiguous. Its dyadic scale structure produces the Mellin denominator $1-2^{-z}$ and hence the periodic correction.

Newton iteration. Newton iteration solves $h(x) = 0$ by $x_{n+1} = x_n - h(x_n)/h'(x_n)$. Near a simple root it converges quadratically under standard hypotheses. Stable endpoint evaluation of inverse Fabius values or saddle coordinates can use Newton's method with monotone bracketing and exact asymptotic initial guesses. The lower-Lambert equation $\lambda 2^{-\lambda} = x$ may also be solved numerically this way when a special-function implementation is unavailable.

Noise floor. The numerical noise floor is the error level below which rounding, cancellation, underflow, or implementation effects dominate the signal being measured. The Fabius periodic correction has very small amplitude, so ordinary floating-point evaluation can obscure it. Exact rational endpoint values, arbitrary precision, and subtraction of the full nonoscillatory main term are required before interpreting residual plots. A flat residual at the noise floor does not establish absence of a mathematical term.

Nonconstant periodic correction. A nonconstant periodic correction is a periodic term with at least one nonzero Fourier mode. In the Fabius endpoint asymptotic, every relevant Gamma–zeta coefficient is explicit, and nonvanishing proves that the correction cannot be absorbed into a constant. Its numerical amplitude is extremely small because Gamma values decay rapidly on vertical lines, but asymptotic correctness depends on logical size relative to the stated remainder, not on visual prominence.

Normal convergence. A series of functions converges normally on a domain if it converges absolutely and uniformly on every compact subset, usually by the Weierstrass M-test. Normal convergence of holomorphic functions permits termwise differentiation and yields a holomorphic sum. Infinite transform products and Fourier series of the smooth periodic correction are controlled in this manner. The term is stronger than pointwise convergence and is especially useful for complex-analytic families.

Normalized saddle mass. The normalized saddle mass is what remains of a saddle integral after factoring out the exponential at the saddle, the leading amplitude, and the Gaussian normalization. It tends to one and has a formal expansion $1 + \sum_{j\geq 1} a_j \lambda^{-j}$. This normalization isolates universal Gaussian moments from problem-specific phase jets. Its logarithm supplies the additive all-orders corrections A_j in the Fabius endpoint formula.

Nowhere analytic. A smooth function is nowhere analytic on a set if it is not real analytic at any point of that set. This is stronger than having isolated nonanalytic points. The Fabius function is nowhere analytic on its nonconstant interval: dyadic points are dense, their formal Taylor series terminate, and if the function were analytic near any point, analytic continuation and overlap with such a dyadic neighborhood would force local polynomial behavior incompatible with strict shape and scaling. The exterior constant tails remain analytic away from the endpoints.

Null set. A null set is a set of measure zero. Altering an integrable function on a null set does not change its Lebesgue integral or its L^p equivalence class, but it can change pointwise properties such as continuity or exact functional equations. Formalizations that reason with continuous representatives cannot silently replace functions almost everywhere. The Fabius theory usually works with canonical continuous or smooth versions rather than equivalence classes.

Numerator sequence. A numerator sequence records the integer numerators obtained after multiplying rational quantities by a prescribed denominator. The choice of denominator is part of the definition and can be designed to expose recurrence, parity, or divisibility. The Fabius directory contains several such sequences for moments, half moments, and dyadic endpoint values. Similar names in different papers need a notation crosswalk because their normalizations are not interchangeable.

Numerical residual. A numerical residual is the discrepancy obtained by substituting a computed approximation into an equation or by subtracting a predicted approximation from trusted data. A small residual is evidence, not by itself a proof, because conditioning and cancellation matter. Exact rational dyadic values provide unusually strong reference data for testing the corrected Fabius asymptotic. Comparing residuals across scales reveals the periodic oscillation and the order of the first omitted saddle term.

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15 O

Odd cosine series. An odd cosine series is a cosine expansion containing only odd harmonics $\cos((2m+1)\pi x)$. It arises when even-frequency Fourier coefficients vanish. For the up-function, zeros of the sinc product force this pattern in the chosen periodization. Despite the adjective odd, the resulting function is even in x; odd refers to the harmonic indices, not parity of the function.

One-periodic function. A function P is one-periodic when $P(u+1) = P(u)$ for all u. It is determined by its values on any interval of length one and has Fourier frequencies in $\mathbb{Z}$. The endpoint correction and every all-orders coefficient A_j in the Fabius logarithmic expansion are one-periodic functions of the lower-Lambert coordinate. Centering sets the mean of the leading correction to zero.

Operation count. An operation count estimates the number of arithmetic or logical steps performed by an algorithm as a function of input size. For integer inputs, bit length and Hamming weight are more informative than numerical magnitude. The fast dyadic Fabius evaluator makes one main recursive reduction per set bit and evaluates finite Taylor blocks with bounded-degree arithmetic. Exact bit complexity also depends on growth of numerators and denominators, so a scalar operation count is only one layer of analysis.

Order of vanishing. A function has a zero of order m at x_0 if its first $m-1$ derivatives vanish there and the m-th derivative does not. Infinite-order vanishing is flatness. Finite Thue–Morse generating polynomials have high-order zeros at $z=1$, encoding Prouhet cancellation of low powers. At an interior dyadic point, the Fabius Taylor polynomial has exact highest degree determined by the first nonzero scaled integer value, not by a zero of the function itself.

Ordered composition. An ordered composition of n is a sequence of positive integers summing to n. It indexes terms produced by expanding powers of a series or taking a formal logarithm. The explicit Fabius saddle mass coefficients sum over compositions of $2j$, with additional subset choices selecting phase or amplitude contributions; the logarithmic coefficients sum over compositions of j. Ordered indexing avoids symmetry factors but produces more terms than partition notation.

Ordinary generating function. The ordinary generating function of (a_n) is $A(z)=\sum_{n\geq 0} a_n z^n$. Its product corresponds to ordinary discrete convolution. The finite Thue–Morse generating polynomial factors digitwise, and iterated prefix sums correspond to division by powers of $1-z$ in a formal setting. Ordinary and exponential generating functions occur side by side in the repository and encode different convolution laws.

Orthogonal polynomial. A sequence (p_n) is orthogonal with respect to a measure μ if $\int p_n p_m \, \mathrm{d}\mu = 0$ for $m \neq n$, with $\deg p_n = n$. Orthogonal polynomials satisfy three-term recurrences and provide spectral bases for approximation. Legendre polynomials are orthogonal for unit weight on $[-1,1]$ and are used to expand the up-function. Orthogonality makes coefficient extraction and least-squares optimality immediate.

Orthogonal projection. The orthogonal projection of a Hilbert-space vector f onto a closed subspace V is the unique $p \in V$ with $f-p$ orthogonal to V. It minimizes $\|f-p\|$. The degree-n Fourier-Legendre partial sum is the orthogonal projection onto polynomials of degree at most n. The corresponding Pythagorean identity quantifies exactly how the squared error decreases when another Legendre coefficient is included.

Orthogonality. Vectors or functions are orthogonal when their inner product is zero. In $L^2[-1,1]$, $\langle f, g\rangle = \int f\overline{g}$. Orthogonality eliminates cross terms in norm calculations and yields the Pythagorean identity. The Legendre basis separates the polynomial components of the up-function, while Fourier exponentials separate periodic modes of the endpoint correction.

Oscillatory correction. An oscillatory correction varies rather than tending monotonically to a constant, often through sine, cosine, or a periodic function of a slowly varying phase. Its mean may be zero while its pointwise effect persists indefinitely. The Fabius correction is smooth and one-periodic in $\lambda(x)$, so sequences approaching the endpoint through different phases have different normalized limits. The term is asymptotically indispensable despite its tiny amplitude.

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16 P

Paley–Wiener theorem. A Paley–Wiener theorem relates compact support to holomorphic extension and exponential type of the Fourier transform. In the repository's convention, a smooth function supported in $[-A, A]$ has an entire transform that decreases faster than every power on the real axis and grows no faster than an exponential controlled by $2\pi A|\operatorname{Im} z|$; converses require the corresponding analytic bounds. This theorem explains the compatibility of the up-function's compact support with its entire sinc product. The repository's support proof itself uses the probabilistic convolution construction, so the glossary does not claim that rapid real-axis decay alone implies compact support.

Parity. Parity distinguishes even integers from odd integers. It is additive modulo two and governs the Thue–Morse sign through the parity of binary digit sum. Parity assertions about normalized Fabius moment numerators are stronger than nonvanishing and determine exact 2-adic valuations. In finite q-binomial sums, parity also controls sign patterns and even-odd term separation.

Parity-power series. The parity-power series is a signed global Fabius representation whose summands are organized by binary parity data and powers arising from finite Taylor blocks. A corrected version in the repository includes the precise zeroth term and domain convention needed for global validity. It converges absolutely. The name is local to the development and should be understood through its displayed summands rather than as a standard class of power series.

Parseval identity. Parseval's identity equates the L^2 norm of a function with the sum of squared coefficients in a complete orthonormal basis. For Fourier series it is $\|f\|_2^2 = \sum |c_k|^2$; for Legendre series the norm factors reflect polynomial normalization. It justifies energy estimates and coefficient square summability. In the Fabius context it complements the stronger absolute-uniform convergence results for the up-function expansion.

Partition of unity. A partition of unity is a family of nonnegative functions whose sum equals one, often locally finite and subordinate to a covering. Translates of the up-function satisfy exact partition identities because the Fourier transform vanishes at nonzero integers under an appropriate normalization. Scaled versions yield further partitions. These identities are analytically analogous to cardinal spline partitions and are useful in approximation and sampling.

Periodic correction. A periodic correction is a periodic function added to an otherwise smooth asymptotic main term to account for discrete-scale effects. In the Fabius endpoint theory it arises from the nonzero imaginary poles of $(1 - 2^{-s})^{-1}$ in Mellin space. Its centered form has zero mean and explicit Gamma–zeta Fourier coefficients. Uniqueness means that no different continuous one-periodic function can satisfy the same sharp remainder estimate.

Periodic correction Psi. The symbol Ψ denotes the centered smooth one-periodic function in the corrected Fabius endpoint expansion. Its mean over one period is zero, and its nonzero Fourier coefficients are explicit Gamma–zeta values. It is evaluated at the exact lower-Lambert coordinate $\lambda(x)$. The centering convention moves the zero Fourier mode into the endpoint constant, making Ψ unique under the stated sharp remainder.

Periodic mean. The mean of a one-periodic integrable function P is $\int_0^1 P(u)\,\mathrm{d}u$. It is the zero-frequency Fourier coefficient. Subtracting it centers the function; adding it to the constant term keeps the full expression unchanged. Repository modules compute the mean of the raw Mellin periodic term so that the endpoint constant and centered oscillation have a canonical normalization.

Periodicity. A function has period $T > 0$ if $f(x+T) = f(x)$. The smallest positive period, when it exists, is the fundamental period. Periodicity can arise from translation symmetry or from multiplicative scaling after a logarithmic change of variables. The Fabius asymptotic uses period one in the lower-Lambert coordinate, corresponding to the base-two structure of the original equation.

Perturbation expansion. A perturbation expansion expresses a solution or integral as a formal or asymptotic series in a small parameter. One expands the phase, amplitude, saddle location, or recurrence around a limiting model and solves coefficient equations successively. The inverse-saddle parameter $1/\lambda$ is the perturbation parameter in the Fabius all-orders analysis. Formal coefficient algebra and analytic remainder estimates are separate components of a valid perturbation proof.

Phase function. The phase function is the exponent controlling an oscillatory or Laplace-type integral, commonly $\Phi(z)$ in $\int A(z)\mathrm{e}^{\lambda\Phi(z)}\,\mathrm{d}z$. Saddles satisfy $\Phi'(z) = 0$; the real part of Φ determines concentration and decay. In the Fabius Bromwich integral the phase incorporates the linear inversion term and the logarithm of the negative-Laplace product. The dominant reference balance motivates the lower-Lambert coordinate; the formal saddle-coordinate module records its exact algebraic identities without asserting that it solves the full phase's exact critical-point equation.

Piecewise polynomial. A piecewise-polynomial function agrees with a polynomial on each interval of a finite or locally finite partition. Splines impose smoothness conditions at the knots. Finite convolutions of uniform densities are piecewise polynomial, and their degrees increase with the number of factors. The centered Fabius approximants are therefore exactly evaluable on each dyadic cell, which is crucial for primitive-recursive computation.

Pochhammer symbol. The ordinary Pochhammer symbol $(a)_n = a(a+1)\cdots(a+n-1)$ is the rising factorial, with $(a)_0 = 1$. It appears in hypergeometric series and Gamma ratios: $(a)_n = \Gamma(a+n)/\Gamma(a)$. The q-Pochhammer symbol is a different multiplicative object $(a;q)_n = \prod_{j=0}^{n-1}(1-aq^j)$. The Fabius q-calculus mainly uses the latter, so the base parameter should always be displayed.

Pointwise convergence. A sequence f_n converges pointwise to f if $f_n(x) \to f(x)$ for each fixed x. The index after which an approximation is accurate may depend on x, unlike uniform convergence. Pointwise convergence alone does not preserve continuity or justify interchange with integration. The repository's stronger spline and Legendre results supply uniform convergence, while some discrete-limit formulas are first established pointwise and then strengthened through bounded rows or compactness.

Poisson summation. Poisson summation relates samples of a function to samples of its Fourier transform: under suitable decay, $\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{k\in\mathbb{Z}} \widehat{f}(k)\mathrm{e}^{2\pi\mathrm{i}kx}$. If all nonzero integer transform samples vanish, the periodized sum is constant. Applying this to the up-function's sinc product yields exact partitions of unity and sampling identities. Smooth compact support supplies the regularity needed for classical justification.

Pole. A pole of order m at z_0 is an isolated singularity where $(z-z_0)^m f(z)$ extends holomorphically and nonvanishingly. The Laurent series has finitely many negative powers. Mellin contour shifts collect residues at poles; a real pole generates polynomial-logarithmic main terms, while imaginary translates generate periodic modes. The dyadic factor $(1-2^{-s})^{-1}$ creates the relevant arithmetic lattice of poles.

Polynomial approximation. Polynomial approximation seeks polynomials close to a function in a chosen norm. Weierstrass' theorem gives uniform approximation for continuous functions on compact intervals, while orthogonal projections give least-squares approximants. The up-function's Fourier-Legendre series provides explicit polynomial approximants with exact coefficients and proven absolute uniform convergence. This is distinct from the finite spline approximation, whose pieces are polynomial but whose global approximant is not a single polynomial.

Polynomial reproduction. An approximation or summation scheme reproduces polynomials up to degree m if it returns every such polynomial exactly. Finite-difference annihilation and partition-of-unity identities are dual manifestations of reproduction. Gaussian rows on geometric nodes reproduce the leading-coefficient functional after annihilating lower degrees. In spline and Toeplitz limits, exact mass and low-moment conditions control the approximation order.

Power series. A power series $\sum a_n(z-z_0)^n$ converges inside a disk determined by its radius of convergence and defines a holomorphic function there. Within the disk it may be differentiated and integrated termwise. A formal power series has the same algebra but no convergence claim. The Fabius function's Taylor series at dyadic points is finite yet fails to represent the function locally, whereas its transform power series is entire and analytically valid.

Primitive-recursive function. Primitive-recursive functions are built from zero, successor, and projections using composition and primitive recursion. They form a robust class of total computable functions with explicit structural termination, though they do not include every computable function. The repository expresses the finite-spline evaluator, signed-natural arithmetic, precision selection, and continuity modulus in primitive-recursive terms. This supplies a constructive proof of Fabius computability stronger than an informal numerical algorithm.

Principal branch. A principal branch is a conventional single-valued choice from a multivalued complex function, such as the principal logarithm or W_0 branch of Lambert W. Branch choices determine cuts, arguments, and continuity domains. The Fabius lower-Lambert reference coordinate uses the nonprincipal real branch W_{-1}; the principal branch selects a bounded root and therefore the wrong endpoint scale. Complex shifts and logarithms in transform formulas likewise require consistent branch declarations.

Probabilistic representation. A probabilistic representation identifies a deterministic function or sequence with a probability, expectation, density, or law. It can make positivity, monotonicity, normalization, and convolution structure transparent. The Fabius function is the CDF of an infinite dyadic weighted sum of independent uniforms; the up-function is the density of a centered affine version. Transform products and finite spline approximants follow directly from this representation.

Probability density. A probability density f is a nonnegative integrable function with total integral one; probabilities are $\mathbb{P}(X \in A) = \int_A f(x)\ \mathrm{d}x$, and its CDF is $F(x) = \int_{-\infty}^{x} f(t)\ \mathrm{d}t$. For $X = \sum_{k\geq 1} 2^{-k} U_k$, the bounded Fabius function is $F(x) = \mathbb{P}(X \leq x)$ and

$$f_X(x) = F'(x) = 2\,\mathrm{up}(2x-1).$$

Equivalently, $Y = 2X - 1$ has density up. Strict positivity on the support interior gives strict monotonicity of the CDF.

Probability measure. A probability measure is a countably additive measure of total mass one. It provides the abstract framework for random variables and distributions. Pushforwards, products, and convolutions construct new laws. The infinite product probability space carrying independent uniform variables and the pushforward under their weighted sum gives a rigorous construction of the Fabius measure.

Product measure. Given probability spaces (Ω_n, μ_n), their product measure is the canonical measure on the Cartesian product under which coordinate random variables are independent and have laws μ_n. Infinite product measures are constructed from consistent finite-dimensional distributions. The Fabius random series can be defined on a product of unit intervals, with each coordinate carrying a uniform law. Absolute summability of weights makes the sum measurable and convergent.

Prouhet cancellation. Prouhet cancellation is the vanishing of signed power sums generated by the Thue–Morse sequence. For a complete block of length 2^m, $\sum_{n=0}^{2^m-1} \varepsilon_n n^r = 0$ for $0 \leq r < m$. It follows from the order-m zero at $z = 1$ of $\prod_{j<m}(1 - z^{2^j})$. This annihilation removes lower-degree polynomial contributions in dyadic Fabius sums and underlies the exact leading-coefficient extraction.

Pushforward measure. For a measurable map $T : X \to Y$ and a measure μ on X, the pushforward $T_{\#}\mu$ is defined by $T_{\#}\mu(A) = \mu(T^{-1}(A))$. It is the law of $T(X)$ when X has law μ. The Fabius measure is the pushforward of an infinite product of uniform measures under the weighted-sum map. Affine pushforwards relate the bounded distribution and centered up-density.

Pythagorean identity. In an inner-product space, if $u \perp v$ then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$. For an orthogonal projection p of f, one has $\|f-q\|^2 = \|f-p\|^2 + \|p-q\|^2$ for every q in the subspace. This proves least-squares optimality and quantifies the error reduction of Fourier-Legendre truncation for the up-function.

17 Q

q-binomial coefficient. The q-binomial coefficient is the Gaussian coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q;q)_n/((q;q)_k(q;q)_{n-k})$. It satisfies a q-Pascal recurrence and counts subspaces of a finite vector space when q is a prime power, while algebraically it is a polynomial in q. The Fabius directory uses the analytic value $q = 1/2$, where a normalized row gives interpolation and Toeplitz weights. The parameter is a scalar base, not a probability.

q-binomial scalar extension. Scalar extension transports a polynomial or finite-sum identity proved over $\mathbb{Q}$ or $\mathbb{R}$ into a larger field such as $\mathbb{C}$. Since Gaussian binomial coefficients at $q = 1/2$ are rational, their identities can be mapped through the canonical embedding. The repository separates this algebraic step from analytic convergence, allowing the complex-shift q-binomial Taylor identity to reuse an exact real polynomial theorem.

q-binomial theorem. The finite q-binomial theorem states $(z;q)_n = \sum_{k=0}^{n}(-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k$. Variants follow by changing signs or powers. At $q = 1/2$, evaluations at $z = \pm 1/2$ prove that the discrete-limit rows have mass one and uniformly bounded total variation. Coefficient comparisons also give the geometric-node finite-difference identities used in exact dyadic evaluation.

q-calculus. q-calculus is a deformation of ordinary calculus in which additive increments are replaced by multiplicative scaling by q. Its basic objects include q-integers, q-factorials, q-differences, Gaussian binomial coefficients, and q-Pochhammer symbols. The Fabius problem has a natural half-base $q = 1/2$ because every differential scaling step doubles the argument. Here q-calculus is finite and algebraic rather than a replacement for the ordinary derivative defining F.

q-discrete limit. The complex q-discrete limit is a sequence of finite Gaussian-binomial combinations of complex-shift spline approximants that converges to the signed global Fabius extension. After reindexing, each approximant is a Toeplitz row with total mass one and uniformly bounded variation, sampling only sufficiently advanced splines. A finite row can depend on the complex shift Q; the limit is independent of fixed Q. This limit is not termwise identical to the global q-series even though both converge to the same function.

q-factorial. The q-integer is $[n]_q = (1-q^n)/(1-q)$, and the q-factorial is $[n]_q! = \prod_{j=1}^{n}[j]_q$. It is related to the q-Pochhammer product by $[n]_q! = (q;q)_n/(1-q)^n$. Gaussian binomial coefficients are ratios of q-factorials. At $q = 1/2$, clearing powers of two converts these products into odd Mersenne products.

q-finite difference. A q-finite difference compares values at multiplicatively related points, often through $D_q f(x) = (f(x) - f(qx))/((1-q)x)$, or through a finite Gaussian-weighted node sum. The repository mainly uses the latter leading-coefficient functional on geometric nodes. Like an ordinary n-th difference, it annihilates polynomials of degree below n, but the weights and nodes encode powers of two.

q-integer. The q-integer $[n]_q = 1 + q + \cdots + q^{n-1} = (1-q^n)/(1-q)$ deforms the ordinary integer n, recovered as $q \to 1$. Products of q-integers form q-factorials. Although the Fabius formulas are often written directly with $(q;q)_n$ and Gaussian coefficients, q-integers explain the parallel with ordinary binomial algebra and the appearance of factors $2^j - 1$ at half base.

q-Pochhammer symbol. The finite q-Pochhammer symbol is $(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j)$, with $(a;q)_0 = 1$; the infinite version is $(a;q)_\infty = \prod_{j\geq 0}(1 - aq^j)$ when it converges. At $a = q = 1/2$, multiplying by a suitable power of two yields $\prod_{j=1}^{n}(2^j - 1)$. These products normalize Gaussian rows, dyadic formulas, and denominator sequences throughout the Fabius arithmetic.

q-product. A q-product is a finite or infinite product with geometrically progressing factors, typically a q-Pochhammer symbol. Its zeros and ratios encode q-recurrences and basic hypergeometric coefficients. The Fabius half-base products link geometric interpolation denominators to Mersenne factors. They should be distinguished from the independent sinc product of the Fourier transform, although both reflect the same dyadic scale hierarchy.

q-series. A q-series is a series whose coefficients or exponents are organized by powers of a base q, often using q-Pochhammer symbols. It may be formal, terminating, or analytically convergent. The global Fabius q-binomial series is an absolutely convergent sum of finite Taylor blocks expressed through Gaussian rows and a fixed complex shift. The convergence comes from binary reduction estimates, not merely from formal q-series algebra.

q-Taylor identity. The repository's q-Taylor identity states that a finite Gaussian-binomial expression with arbitrary common complex shift Q equals a real Taylor block embedded in $\mathbb{C}$. Translation invariance follows because the order-n geometric finite difference annihilates every lower-degree term generated by expanding the shift. This identity converts local derivative data into the finite summands of the global complex q-series.

Quadratic logarithmic law. A quadratic logarithmic law states that the leading behavior of $\log F(x)$ is a negative constant times $(\log(1/x))^2$. It is a coarse consequence of the sharp endpoint expansion and ignores nested logs and periodic fluctuations. Such a law proves superpolynomial flatness and gives the correct broad decay class. It is insufficient for asymptotic equivalence or accurate numerical evaluation because lower terms in the exponent remain large or bounded but nonconvergent.

Quotient-of-exponentials approximation. A quotient-of-exponentials approximation replaces a complicated product, recurrence, or sum by a ratio of exponential expressions suggested by leading logarithms. Such an ansatz can capture a coarse scale while missing a linear saddle residue, a periodic factor, or a normalization constant. Repository modules named for quotient-exponential mismatch formalize why certain attractive informal proxies cannot achieve the claimed error. Substitution into the exact stationary equation is used to expose the residual term.

18 R

Radius asymptotic. A radius asymptotic describes the size of the neighborhood around a saddle on which a local expansion is used, often as a function of the large parameter. The radius must grow enough to capture Gaussian mass but remain small enough that higher phase terms are controlled. Optimizing this tradeoff is part of a quantitative saddle proof. The Fabius modules derive admissible central radii and show that tail and Taylor remainders meet the target inverse-λ order.

Random series. A random series is a series of random variables. Its convergence can be studied almost surely, in probability, or in L^p. The Fabius random series has bounded independent terms with geometrically summable weights, so it converges absolutely for every outcome. Its distributional self-similarity under removal of the first summand leads to the fixed-point and differential equations.

Rapid decay. A function $g(t)$ has rapid, or Schwartz, decay if $|t|^N g(t) \to 0$ for every N, usually together with analogous bounds for derivatives. The Fourier transform of a compactly supported C^∞ function decays rapidly along the real axis. This property ensures absolute convergence of inversion integrals and Fourier coefficient series. The explicit sinc product supplies a direct route to rapid-decay estimates for the up-function transform.

Rate of convergence. A rate of convergence quantifies how fast an error tends to zero, for example geometrically, polynomially, or as an inverse logarithm. A certified rate provides a function mapping tolerance to a sufficient index. Fabius fixed-point iterates converge geometrically, centered splines have a dyadic uniform rate, Fourier-Legendre coefficients decay rapidly, and the endpoint saddle expansion has inverse-Lambert orders. These rates serve different algorithms and should not be compared without matching cost models.

Rational recurrence. A rational recurrence determines each term using rational operations on earlier terms. Triangular moment recurrences are typical. They prove rationality inductively but may obscure integrality and denominator growth. The Fabius arithmetic converts selected rational recurrences into division-free integer recurrences by multiplying through canonical factors, allowing parity and valuation proofs unavailable from the raw recurrence alone.

Rationality. A quantity is rational when it belongs to $\mathbb{Q}$. Rationality at all dyadic arguments is a remarkable property of the Fabius function, despite its definition through an infinite random series or transform product. It follows from terminating Taylor data and explicit finite Thue–Morse/moment formulas. Rationality alone does not identify the reduced denominator; the valuation and common-denominator theorems refine it.

Real-analytic function. A real function is analytic at x_0 if it agrees with a convergent power series on some neighborhood of x_0. Real analyticity is stronger than C^∞ smoothness and entails local determination by derivatives. The bounded Fabius function is analytic precisely on $\mathbb{R} \setminus [0, 1]$ and nowhere analytic at every point of $[0, 1]$. The up-function is analytic precisely on $\mathbb{R} \setminus [-1, 1]$ and nowhere analytic on the entire closed support. Thus the flat transition endpoints belong to the nowhere-analytic sets rather than to the analytic constant-tail regions.

Reciprocal-power table. A reciprocal-power table is a list of exact values, normalized numerators, or asymptotic data for $F(2^{-n})$. These samples sit at the natural dyadic endpoint scales and are linked directly to half moments. The table is useful for discovering parity patterns, checking denominator formulas, and validating the periodic saddle asymptotic. Exact rational entries avoid the catastrophic underflow that affects direct floating-point evaluation for large n.

Recurrence relation. A recurrence relation defines terms of a sequence or family from earlier terms or smaller parameters. It is triangular when the new term appears with an invertible coefficient and only lower indices occur elsewhere. Recurrences can be linear or nonlinear, homogeneous or inhomogeneous, scalar or polynomial-valued. The Fabius directory uses them for moments, numerators, Bernoulli coefficients, saddle jets, Lambert coefficients, and exact dyadic evaluation.

Recursive fast name. A recursive fast name is a computable sequence of rational approximations whose error decreases at a prescribed exponential rate. Recursive means there is an effective algorithm for every precision, while fast refers to the convergence guarantee. The signed-natural dyadic representation in the repository is a concrete fast name. Composition with effective arithmetic and the finite-spline evaluator yields names for Fabius values.

Recursive modulus. A recursive modulus is a computable rate function for continuity, convergence, or approximation. Given a desired output accuracy, it returns a sufficient input accuracy or truncation level. The Fabius computability theorem provides both a uniform-continuity modulus from the Lipschitz constant and an approximation modulus from the centered spline error. Together they make the qualitative algorithm uniform over real inputs.

Reference tail. A reference tail is a simpler majorant or comparison integral used to bound the remote part of a contour integral. Its decay is chosen to dominate the true integrand after the central saddle region, while remaining explicitly integrable. Fabius saddle modules separate reference-tail estimates from local phase algebra, allowing the proof to show that omitted contour portions are below every retained inverse-saddle order.

Reference weight. A reference weight is a normalized model density, usually Gaussian, against which the local saddle integrand is compared. Moments of the reference weight generate universal double-factorial coefficients. The true normalized integrand is expanded as the reference weight times phase and amplitude corrections. Controlling the ratio on the central region and the tails produces both the leading mass and all higher saddle coefficients.

Refinement equation. A refinement equation expresses a function in terms of scaled and translated copies of itself, typically $f(x) = \sum_k c_k f(2x - k)$. It is fundamental in wavelets, splines, and subdivision schemes. The up-function satisfies a differential/refinement identity derived from the Fabius equation; its Fourier transform then satisfies an infinite product relation. Support and normalization select the canonical compactly supported solution.

Refinement operator. A refinement operator maps a function to a combination or integral of its dilates and translates. Fixed points solve the associated refinement equation. Depending on the norm and normalization, the operator may be a contraction or preserve mass and positivity. Iterating the Fabius refinement operator produces increasingly fine probability or spline approximants and gives a constructive existence proof.

Reflection symmetry. Reflection symmetry about c means $f(c+x) = f(c-x)$ for an even-type function, or $F(c + x) = 1 - F(c - x)$ for a symmetric CDF. The bounded Fabius function satisfies $F(1 - x) = 1 - F(x)$, while the up-function is even about zero. Symmetry halves many integrals, kills odd moments or Legendre coefficients, and transfers endpoint and convexity results from one side to the other.

Relative error. The relative error of an approximation g to a nonzero quantity f is $|g - f|/|f|$. For extremely small Fabius endpoint values, a tiny absolute error can still be a huge relative error. Logarithmic asymptotics are often analyzed first because direct values underflow, but exponentiating a bounded missing term causes an order-one relative oscillation. Exact rational benchmarks make relative-error assessment reliable.

Relative saddle mass. Relative saddle mass is the ratio of the exact localized saddle integral to its leading Gaussian approximation. It tends to one and admits an inverse-saddle expansion. Expressing corrections relatively avoids repeatedly carrying the dominant exponential and square-root prefactors. The repository's mass-all-orders modules compute this ratio first and then take its formal logarithm to obtain the endpoint expansion for $\log F$.

Remainder estimate. A remainder estimate bounds the error after truncating a series, Taylor expansion, contour localization, or asymptotic formula. A formal coefficient identity becomes an analytic theorem only when its remainder is controlled uniformly in the claimed regime. The Fabius development has separate modules for Taylor remainders, saddle central errors, minor arcs, tail flatness, and inverse-Lambert transfer. Their combination certifies each stated $\mathcal{O}$-term.

Removable singularity. A removable singularity is an isolated point where a function is initially undefined or apparently singular but has a finite limit and extends holomorphically or continuously. The sinc function $\sin z/z$ has the removable value 1 at $z = 0$. Regularized Bose kernels likewise subtract a principal singular term to produce a smooth remainder, though the original integral may still require a finite part. Explicitly assigning removable values avoids domain defects in formal statements.

Representation independence. Representation independence means that a mathematical output depends only on the represented object, not on a nonunique encoding. Dyadic rationals can be written at multiple levels, real numbers can have two binary expansions, and signed dyadic numerators can have redundant positive-negative pairs. Exact Fabius evaluators prove compatibility under refinement and coding equivalence. This is necessary before an implementation can be regarded as a function on rationals or reals rather than on syntax.

Reshetnikov integers. For $n \geq 1$, the repository defines

$$R_n = 2^{\binom{n-1}{2}}(2n)!\,F(2^{-n})\prod_{j=1}^{\lfloor n/2 \rfloor}(2^{2j}-1).$$

The main arithmetic theorem proves that every R_n is a positive odd integer. This exact normalization contains a factorial, a triangular power of two, and odd Mersenne factors, but no double factorial. Its integrality refines earlier conjectural observations and packages the cancellation in inverse-dyadic values into a natural integer sequence.

Residue. The residue of a meromorphic function at z_0 is the coefficient of $(z - z_0)^{-1}$ in its Laurent expansion. The residue theorem converts a contour shift into the sum of residues of crossed poles. In Mellin analysis, residues produce asymptotic terms. For the Fabius dyadic harmonic sum, residues at the imaginary lattice yield the Fourier coefficients of the periodic correction.

Riemann zeta function. The Riemann zeta function is $\zeta(s) = \sum_{n\geq 1} n^{-s}$ for $\Re s > 1$, continued meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. It appears in Mellin transforms of Bose-type kernels and in Euler–Maclaurin constants. The Fabius periodic Fourier coefficients use $\zeta(1 - \chi_k)$ at nonreal points. Nonvanishing on the line $\Re s = 1$ at these frequencies helps establish nonconstancy.

Rodrigues formula. Rodrigues' formula constructs Legendre polynomials by $P_n(x) = \frac{1}{2^n n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2 - 1)^n$. It proves degree, parity, endpoint normalization, and orthogonality after integration by parts. In the up-function Legendre analysis, it converts coefficients into integrals involving derivatives or moments, which can then be evaluated from the Fabius recurrence data.

Rounding to nearest dyadic. Rounding to the nearest dyadic at precision p selects an integer a so that $|x - a/2^p| \leq 2^{-p-1}$, with a specified tie convention. Combined with a Lipschitz bound, it converts input approximation error into output error. The computable Fabius algorithm can round a real name to a dyadic grid, evaluate a finite spline exactly there, and add the certified spline and input-propagation errors.

Rvachev up-function. Rvachev's up-function is the even compactly supported bump $\mathrm{up}(x) =$

$F(1-|x|)$ obtained from the bounded Fabius function. It is nonnegative, supported exactly on $[-1,1]$, normalized by $\mathrm{up}(0)=1$, strictly increasing on $[-1,0]$, and strictly decreasing on $[0,1]$. It is the centered density form of the same probability law, has an entire sinc-product Fourier transform, satisfies refinement identities, and is a canonical atomic function.

19 S

Saddle coefficient recurrence. A saddle coefficient recurrence computes the coefficients of a local integral expansion from lower-order phase and amplitude data. Gaussian parity eliminates odd total powers, while each surviving coefficient is a finite combination of jets and double-factorial moments. The repository supplies recurrences suitable for Lean induction as well as explicit closed forms indexed by ordered compositions and subsets. The logarithmic coefficients are then obtained from the mass coefficients by a formal-log recurrence.

Saddle jet. A saddle jet is the finite list of derivatives of the phase or amplitude at the saddle, usually normalized by powers of the large parameter so that each component has a finite limit or periodic dependence. These jets determine every fixed order of the steepest-descent expansion. Repository modules derive closed forms for the Fabius saddle jets using Stirling-number algebra and the periodic Mellin correction, then feed them into generic coefficient machinery.

Saddle mass. The saddle mass is the integral contribution from the central neighborhood after the dominant exponential has been extracted. Its leading term is Gaussian, proportional to the inverse square root of the phase curvature. Higher corrections account for nonquadratic phase terms and amplitude variation. In the Fabius problem the normalized saddle mass has an all-orders inverse-λ expansion with periodic coefficients inherited from the negative-Laplace decomposition.

Saddle point. A saddle point of a complex phase Φ is a critical point where $\Phi'(z_0)=0$; along suitable directions the real part decreases and concentrates a contour integral. In the repository's convention the Bromwich contour is parametrized by $z=r+2\pi\mathrm{i}\xi$, while the transform product is evaluated at $-z$. The explicit transform-side reference point is therefore $-r$. For sufficiently small $x>0$, the lower-Lambert quantity $\lambda=-W_{-1}(-x\log 2)/\log 2$ is the reference saddle coordinate, and $r=2^{\lambda}$ satisfies $rx=\lambda$; the formal module uses these exact identities without claiming that λ itself is the contour's exact critical point.

Saddle-point method. The saddle-point method estimates integrals or coefficients with a large parameter by expanding the phase around a critical or asymptotically balanced reference point. A complete proof justifies the central location, evaluates the Gaussian comparison, expands retained corrections, and bounds the tails uniformly. For the Fabius endpoint the repository carries out the corresponding localization, Gaussian comparison, correction, and tail bounds at an explicit lower-Lambert reference radius, extending the calculation to all orders without identifying that radius with an implicit exact critical point. The method is closely related to steepest descent and Laplace's method.

Saddle radius. For sufficiently small $x > 0$, the explicit lower-Lambert reference radius is $r(x) = 2^{\lambda(x)}$, where $\lambda(x) = -W_{-1}(-x \log 2)/\log 2$. It is positive, tends to infinity as $x \downarrow 0$, and obeys the exact identity $r(x)x = \lambda(x)$. In the Bromwich parametrization $z = r + 2\pi i\xi$, the negative-Laplace product is evaluated at the transform-side reference point $-r$. This radius packages the asymptotic saddle coordinates but is distinct from both an implicitly characterized exact critical point and the central-window cutoff.

Sampling identity. A sampling identity relates sums of function values on a lattice to sums of transform values on the reciprocal lattice. For $a > 0$, Poisson summation gives $\sum_{m\in\mathbb{Z}} \mathrm{up}(am) = a^{-1} \sum_{m\in\mathbb{Z}} \widehat{\mathrm{up}}(m/a)$ under the repository's convention. Compact support can reduce the spatial sum to only a few terms for large spacing, while the transform side remains an infinite rapidly convergent series. This yields exact identities for selected up-values. For arbitrary nonzero real spacing the scale factor is $1/|a|$.

Scalar extension. Scalar extension maps an algebraic object over a field or ring into one over a larger scalar system, such as embedding $\mathbb{Q}[x]$ into $\mathbb{C}[x]$. Polynomial identities, finite sums, and recurrences are preserved by ring homomorphisms. The Fabius complex-shift modules prove rational identities once and extend them to complex parameters, keeping analytic convergence arguments separate from algebraic coercions.

Scaling law. A scaling law relates values of a function or distribution under dilation. The global Fabius law $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ and its derivative iterates are exact scaling laws. In transforms, dilation changes frequency inversely and contributes a Jacobian factor. Discrete-scale invariance at ratio two is the common source of dyadic arithmetic, sinc products, and log-periodic endpoint corrections.

Schwartz function. A Schwartz function is C^∞ and, together with every derivative, decays faster than every polynomial: $\sup_x |x^m f^{(n)}(x)| < \infty$ for all m, n. Every compactly supported smooth function is Schwartz. Thus the up-function belongs to the Schwartz space, and its Fourier transform is also Schwartz on the real axis. This guarantees strong inversion and Poisson-summation properties.

Second saddle correction. The second saddle correction is the coefficient of $1/\lambda^2$ in the normalized or logarithmic saddle expansion. It combines phase and amplitude derivatives through higher order and includes cross terms involving the first correction. The repository isolates this coefficient in dedicated modules both as a check on the generic all-orders machinery and as a sharper practical endpoint approximation.

Self-convolution. Self-convolution is $f * f$, representing the sum of two independent variables with density f. A self-similar density may satisfy an exact relation between its self-convolution and a scaled or shifted version of itself. The up-function has such identities inherited from splitting its infinite random series into interlaced dyadic scales. Fourier transforms turn them into elementary product identities.

Self-similarity. Self-similarity means that an object is reproduced, exactly or statistically, after rescaling and possibly translation or averaging. The Fabius random series splits into a first uniform component plus a scaled independent copy of itself. This distributional identity yields the contraction equation, differential scaling, and refinement structure. Because the scale ratio is discrete, the endpoint asymptotic retains periodic information in a logarithmic phase.

Sequential computability. A real function is sequentially computable if it maps every computable input sequence to a computable output sequence by one uniform algorithm. This formulation avoids choosing a particular representation for a single arbitrary real while still enforcing effective dependence on input data. In Grzegorczyk's framework, sequential computability plus effective uniform continuity defines computability of a continuous real function. The finite-spline Fabius algorithm supplies the sequential component.

Sharp asymptotic. A sharp asymptotic includes every contribution needed to attain a specified small remainder, not merely the leading growth order. For $F(x)$ near zero, sharpness requires the exact lower-Lambert phase, the nonconstant periodic correction, the Gaussian prefactor, and stated inverse-saddle errors. A coarser quadratic logarithmic law remains true but is not sharp enough for asymptotic equivalence or first correction terms.

Sharp constant. A constant in an inequality is sharp if it cannot be decreased while preserving the statement. Sharpness is usually proved by exhibiting points or sequences where equality holds or the ratio approaches the constant. The derivative bounds $2^{\binom{m+1}{2}}$ for the Fabius-related functions are sharp because exact dyadic points attain them. The global Lipschitz constant 2 is similarly attained at the midpoint.

Sharp derivative bound. The repository proves $\|F^{(m)}\|_\infty, \|\operatorname{up}^{(m)}\|_\infty, \|\mathcal{F}^{(m)}\|_\infty \le 2^{\binom{m+1}{2}}$, with equality in each case. The bound follows from derivative scaling and the unit bound on the signed extension; equality follows at carefully chosen dyadic arguments. These estimates imply effective flatness, coefficient decay, and stable uniform approximation. They are exact rather than merely exponential-order bounds.

Sharp transfer. A sharp transfer theorem converts asymptotic information for one analytic object into equally precise information for another while preserving constants, periodic phases, and error orders. In the Fabius directory, transfer modules pass from negative-Laplace and saddle estimates to endpoint values, from the continuous endpoint formula to inverse-power sequences, and from the Lambert coordinate to elementary logarithms. The adjective sharp means no term at the target error scale is lost.

Shifted spline. A shifted spline is a spline evaluated after translation, possibly by a complex parameter in a polynomial extension. Real shifts move support and knots; complex shifts are algebraic evaluations of the polynomial pieces rather than probability densities. The Fabius complex-shift spline modules prove convergence for fixed shifts and then combine the approximants with Gaussian q-binomial weights. Common-shift invariance belongs to the finite Taylor functional, not necessarily to each finite spline limit row.

Signed global series. A signed global series represents the smooth global extension $\mathcal{F}$ by summands carrying Thue–Morse or binary-reduction signs. Depending on the presentation, the sum is locally finite or absolutely convergent. On the unit interval it restricts to the bounded Fabius function, but outside it exhibits alternating translated behavior. The signs are structural and cannot be dropped without changing the dilation equation.

Signed measure. A signed measure is a countably additive set function that may take positive and negative values but has finite total variation on the relevant space. It decomposes into positive and negative parts by the Jordan decomposition. Thue–Morse translate sums may be viewed as signed combinations of compactly supported pieces. The signed global Fabius function is not itself a probability CDF, even though it is built from the same local density.

Signed-natural numerator. A signed-natural numerator is a pair $(c^+, c^-) \in \mathbb{N}^2$ representing the integer $c^+ - c^-$. It is redundant because many pairs represent the same integer, but addition, subtraction, and primitive recursion can be implemented using only natural numbers. At dyadic precision p, the represented rational is $(c^+ - c^-)/2^p$. This coding is used in the repository's fast names for computable real sequences.

Sinc function. The sinc function is $\operatorname{sinc} z = \sin z/z$ with removable value 1 at $z = 0$. It is entire and has simple zeros at nonzero integer multiples of π before normalization. The Fourier transform of a centered uniform density is a sinc factor. Multiplying the factors from all dyadic-scale uniforms gives the up-function's entire Fourier product.

Sinc product. The up-function Fourier transform is $\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty} \operatorname{sinc}(\pi z/2^n)$. The product converges locally uniformly and defines an entire function. Its zeros at nonzero integers imply exact periodization and partition-of-unity identities; repeated factors and scale splitting encode convolution laws. Along the real axis, the accumulation of factors gives rapid decay.

Small-argument asymptotic. A small-argument asymptotic describes a function as $x \to 0$. For the Fabius function this regime is singular because all endpoint derivatives vanish, so no nontrivial Taylor series exists. The correct expansion is obtained indirectly through a negative-Laplace transform and a moving saddle. It includes a quadratic logarithmic main scale, nested-log corrections, a lower-Lambert phase, and smooth periodic coefficients.

Smooth compact support. A function has smooth compact support when it is C^∞ and vanishes outside a compact set. Such functions form the test-function space $\mathcal{D}(\mathbb{R})$ and are automatically Schwartz. Their Fourier transforms are entire of exponential type and rapidly decaying on the real axis. The up-function is a highly structured example, distinguished by its refinement equation and exact arithmetic.

Spline. A spline is a piecewise-polynomial function with prescribed continuity of derivatives across knots. Repeated convolution of box functions produces B-splines, whose degree and smoothness increase together. The finite Fabius approximants are explicit centered splines on dyadic knots. They converge uniformly to a C^∞ limit even though each finite approximant has only finite smoothness.

Stable endpoint evaluation. Stable endpoint evaluation computes extremely small Fabius values without losing all relative accuracy. Direct subtraction, naive transform inversion, or standard floating-point arithmetic may underflow or suffer cancellation. Reliable methods use exact dyadic formulas, high-precision logarithmic asymptotics with the periodic phase, or monotone recurrences. The repository discusses operation choice, residual checks, and Newton refinement in a regime where the value itself may be far below machine range.

Stationary point. A stationary point of a differentiable real or complex phase is a point where the first derivative vanishes. It becomes a saddle suitable for asymptotics only when curvature and contour geometry make the contribution dominant. The Fabius audit examines an informal stationary substitution into a Stirling proxy and shows that the exact stationarity equation leaves a linear residual. Thus locating a stationary point does not by itself validate a claimed asymptotic simplification.

Steepest descent. Steepest descent is a contour method in which the path is chosen through a saddle along directions where the imaginary part of the phase is constant and the real part decreases most rapidly. It turns the local integral into a Gaussian model with controlled corrections. The Bromwich saddle method used for Fabius is a specialized steepest-descent analysis adapted to inverse Laplace integration and a dyadic transform product.

Stieltjes constant. The Stieltjes constants γ_n are defined by the Laurent expansion $\zeta(s) = 1/(s-1)+\sum_{n\geq 0}(-1)^n\gamma_n(s-1)^n/n!$, with $\gamma_0 = \gamma$. They appear when Mellin-zeta expressions are expanded near a pole to higher order. Fabius finite-part and all-orders calculations may involve them in constant or derivative coefficients of Gamma–zeta products.

Stirling formula. Stirling's formula approximates the factorial: $n! = \sqrt{2\pi n}(n/\mathrm{e})^n(1+O(1/n))$, or logarithmically $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + O(1/n)$. Explicit Robbins bounds provide a rigorous two-sided error. The Fabius audit uses Stirling's formula both in endpoint sequence asymptotics and to expose a residual linear term missed by an informal stationary substitution.

Stirling numbers. Fix signed first-kind Stirling numbers by

$$x(x-1)\cdots(x-n+1) = \sum_{k=0}^{n} s(n,k)x^k.$$

The saddle-jet weights are not $s(n,m)$, but

$$c(n,m) = [z^m]\prod_{k=1}^{n}(z-k) = s(n+1,m+1), \qquad 0 \leq m \leq n.$$

Hence $c(n,0) = (-1)^n n!$, $c(n,n) = 1$, and $c(n+1,m+1) = c(n,m) - (n+1)c(n,m+1)$. The exact shift is essential: for example $c(2,\cdot) = (2,-3,1)$, whereas $s(n,0) = 0$ for $n \geq 1$.

Strict unimodality. A function is strictly unimodal if it increases strictly up to a unique mode and decreases strictly afterward. Rvachev's up-function is strictly increasing on $[-1, 0]$, attains its unique maximum 1 at zero, and is strictly decreasing on $[0, 1]$. Its derivative signs follow from the positive bounded Fabius function under affine scaling. Strictness strengthens mere symmetry and nonincreasing shape.

Strip of convergence. For a Mellin transform, the strip of convergence is the vertical region $a < \Re s < b$ where the defining integral converges absolutely. Endpoint behavior at zero controls one boundary and behavior at infinity controls the other. Mellin inversion begins on a vertical line inside this strip; asymptotic analysis then uses meromorphic continuation and contour shifting. The regularized Fabius kernels enlarge the workable region by subtracting singular endpoint terms.

Sturm-Liouville equation. A Sturm-Liouville equation has the form $-(p(x)y')' + q(x)y = \lambda w(x)y$ with boundary conditions. Its eigenfunctions are orthogonal with respect to the weight w. Legendre polynomials solve the singular equation $-((1 - x^2)y')' = n(n + 1)y$ on $(-1, 1)$. This differential characterization yields recurrences and integration-by-parts identities used in the up-function's Legendre expansion.

Summability. Summability concerns assigning or proving limits for series, sometimes by methods extending ordinary convergence. In this directory most series are shown to converge in the ordinary sense, often absolutely or uniformly; Toeplitz rows are regular summability transformations carrying convergent sequences to the same limit. The term should not be read as permission to manipulate divergent series without a named regularization such as a finite part.

Support. The support of a continuous function is the closure of the set where it is nonzero. For a measure it is the smallest closed set of full measure, equivalently the points every neighborhood of which has positive mass. The up-function has support exactly $[-1, 1]$, while the bounded Fabius probability law has support $[0, 1]$. Endpoints belong to the support despite zero function values there.

Supremum norm. The supremum norm is $\|f\|_\infty = \sup_x |f(x)|$ on a specified domain. Spaces of bounded continuous functions are complete in this norm, enabling the Banach fixed-point construction. Uniform convergence is convergence in supremum norm. Sharp derivative bounds and finite-spline approximation errors in the Fabius development are stated in this norm.

Symmetry. Symmetry is invariance or a simple covariance under a transformation. The bounded Fabius CDF has complement reflection symmetry about $1/2$, and the up-function has even symmetry about zero. Symmetry determines means, cancels odd moments and coefficients, transfers endpoint formulas, and locates the unique inflection point. It also reduces computations to half of the support.

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20 T

Tail estimate. A tail estimate bounds the contribution from indices or integration regions beyond a cutoff. For infinite random sums it controls the omitted variables; for series it controls the remainder; for contour integrals it controls remote arcs. The dyadic random tail has an explicit deterministic bound, yielding the centered spline sandwich. Saddle tail estimates show that regions outside the central radius are negligible at the desired inverse-λ order.

Tail flatness. Tail flatness refers to rapid vanishing or derivative control in the remote part of a scaled kernel or contour integrand. In the Fabius saddle proof, it supplies uniform bounds when the local Taylor approximation is no longer used. The phrase is repository-specific: it packages estimates showing that the reference tail and true tail differ by less than the target asymptotic remainder.

Taylor block. A Taylor block is a finite Taylor polynomial contribution associated with one dyadic scale in the binary reduction. Because derivatives at a dyadic center vanish above a known order, the block is exact rather than an approximation. Its coefficients are expressed through scaled values of the signed global extension. Gaussian q-binomial identities provide an alternative complex-shift representation of the same block.

Taylor polynomial. The degree-n Taylor polynomial of f at x_0 is $T_n(x) = \sum_{k=0}^{n} f^{(k)}(x_0)(x - x_0)^k/k!$. It approximates f locally when a remainder estimate is available. At reduced dyadic Fabius points, the formal Taylor series terminates exactly at the denominator exponent, yet the polynomial does not equal the function on any neighborhood. This is a striking separation between jet data and analyticity.

Taylor series. The Taylor series of a smooth function at x_0 is the formal series of its derivatives. Analyticity requires that this series converge to the function locally; smoothness alone does not. Fabius endpoint Taylor series are constant because of flatness, and interior dyadic Taylor series are finite polynomials, but neither represents the nonconstant function nearby. Transform Taylor series, by contrast, are entire and valid globally.

Terminating Taylor series. A Taylor series terminates when all derivatives above some order vanish at the center, so the formal series is a polynomial. At a reduced interior point $x_0 = a/2^n$ with $0 < a < 2^n$ and odd a, the Fabius derivatives vanish above order n because repeated dyadic scaling lands outside the active integer pattern. The order-n derivative is nonzero, so the degree is exactly n. Termination does not imply local polynomiality for a merely smooth function.

Termwise differentiation. Termwise differentiation replaces the derivative of a series by the series of derivatives. It is valid under conditions such as locally uniform convergence of the derivative series together with convergence at one point, or normal convergence for holomorphic series. The rapid Fourier and periodic-coefficient convergence in the repository permits repeated differentiation. Formal power-series differentiation is algebraic and requires no analytic hypothesis, so the two settings must be distinguished.

Termwise integration. Termwise integration interchanges a sum and an integral. Tonelli's theorem covers nonnegative terms, Fubini covers absolute integrability, and dominated convergence covers limits of partial sums under a common majorant. Fabius moment, Fourier, and Laplace product derivations use these principles. A merely conditionally convergent signed series requires additional care and is not automatically integrable term by term.

Test-function space. The test-function space $\mathcal{D}(\mathbb{R})$ consists of infinitely differentiable compactly supported functions, equipped with a standard locally convex topology. Its elements are used to define distributions and weak derivatives. The up-function is an explicit member of $\mathcal{D}(\mathbb{R})$. In the original literature it is notable as a highly structured test function satisfying exact refinement and partition identities.

Thue-Morse sequence. The Thue–Morse sequence is the binary automatic sequence $t_n = \mathrm{wt}_2(n) \bmod 2$, satisfying $t_{2n} = t_n$ and $t_{2n+1} = 1 - t_n$. Its sign version is $\varepsilon_n = (-1)^{t_n}$. Finite generating polynomials factor digitwise and have high-order zeros at one, producing Prouhet cancellation. The sequence governs the translate signs of the global Fabius extension and exact dyadic formulas.

Thue-Morse sign. The Thue–Morse sign is $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}$. It obeys $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$, and multiplicative block identities when binary digits do not overlap. Integer values of the signed global Fabius extension are tied to these signs. They also determine the signs of exact highest dyadic derivatives.

Tightness. A family of probability measures on $\mathbb{R}$ is tight if for every $\varepsilon > 0$ some compact set carries mass at least $1 - \varepsilon$ for every measure in the family. Tightness prevents mass from escaping to infinity and is central to weak compactness. Finite Fabius convolution laws have uniformly bounded compact supports, so tightness is immediate. This supports passage to the infinite convolution law.

Toeplitz lemma. A Toeplitz lemma states that a triangular array of weights preserves the limit of a convergent sequence when row sums tend to one, individual fixed-column weights vanish or rows sample sufficiently late terms, and row total variations are uniformly bounded. The repository uses a finite-row variant: after reindexing, all sampled spline degrees are at least the row index, the mass is exactly one, and variation is uniformly bounded by a q-product ratio. Hence the weighted complex approximants converge to $\mathcal{F}$.

Toeplitz row. A Toeplitz row is one row $(\omega_{n,j})_j$ of a triangular summability matrix applied to a sequence. Regular rows have mass tending to one, vanishing influence from fixed early indices, and uniformly bounded total variation. The Fabius complex discrete limit obtains exact row mass and variation from two evaluations of the finite q-binomial theorem. Reindexing also guarantees that every row samples only spline degrees at least n.

Tonelli theorem. Tonelli's theorem allows the order of integration or summation to be exchanged for nonnegative measurable functions, even before finiteness is known. It proves that the iterated integrals and product integral have the same extended value. In probability constructions with nonnegative densities and moments, it is often the cleanest justification. Alternating Thue–Morse sums require Fubini or a separate absolute-convergence argument instead.

Topological support. Topological support is the closed set of points at which an object cannot be locally discarded. For a continuous function it is the closure of its nonzero set; for a measure it consists of points whose every neighborhood has positive measure. The distinction from the set of nonzero values matters at flat endpoints. Thus $\mathrm{up}(\pm 1) = 0$ but $\pm 1 \in \operatorname{supp}\mathrm{up}$.

Total variation. For a signed or complex measure, total variation is the mass of its variation measure. For a finite weight row it is $\sum_j |\omega_j|$, and for a function it is the supremum over partition increments. Uniform total-variation bounds make weighted averaging stable even with alternating coefficients. The q-binomial Toeplitz rows have an explicit uniform bound obtained by evaluating the finite q-binomial theorem at a negative argument.

Transfer theorem. A transfer theorem carries a property or asymptotic from one representation to another. Examples include transferring transform decay to smoothness, saddle estimates to coefficient asymptotics, or endpoint formulas to moment sequences. The Fabius directory contains explicit transfer modules because constants, normalizations, and periodic phases can be lost in an informal passage. A sharp transfer preserves the stated remainder order.

Translate sum. A translate sum has the form $\sum_k c_k f(x-k)$ or a scaled variant. With compact support it may be locally finite. Constant positive coefficients can form partitions of unity; Thue–Morse signs create a signed global extension. Fourier transformation converts translation into multiplication by phases, making periodicity and cancellation visible.

Translated Legendre series. A translated Legendre series expands a function on an interval other than $[-1, 1]$ by composing Legendre polynomials with the affine map to the standard interval. The bounded Fabius function on $[0, 1]$ can be related to the up-function series on $[-1, 1]$ through translation and reflection. Coefficients and convergence must include the Jacobian and normalization of this change of variables.

Triangular cancellation. Triangular cancellation is a layered vanishing mechanism in which lower-degree contributions disappear successively, leaving a top-degree term. In the Fabius dyadic formula, Gaussian interpolation weights annihilate node polynomials below degree n, while Thue–Morse signs independently annihilate lower power sums inside dyadic blocks. The surviving term has a triangular power-of-two normalization. Recognizing both layers explains why apparently complicated finite sums collapse exactly.

Triangular recurrence. A recurrence is triangular when the equation for index n involves the new term with a nonzero coefficient and only terms of smaller index otherwise. It can be solved inductively and often determines uniqueness. Moment, Bernoulli, numerator, and saddle coefficient recurrences in the repository have this form. Clearing denominators may convert a rational triangular recurrence into an integer division-free one.

Triangular reduction. Triangular reduction is the exact decomposition of a dyadic numerator into a large block, a reflected remainder, and lower-level data so that evaluation proceeds inductively. The word triangular reflects both decreasing level and the triangular exponents $\binom{n}{2}$ appearing in normalization. This reduction proves correctness of fast dyadic recursion and exposes binary reflection symmetries.

Truncation error. Truncation error is the difference between an infinite object and a finite partial approximation. It may arise from omitting random summands, series terms, spline levels, or asymptotic corrections. The Fabius finite random tail gives a deterministic support bound and hence a uniform CDF error; Fourier and saddle truncations use summable or exponential estimates. Explicit truncation errors turn formulas into certified algorithms.

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21 U

Unconditional convergence. A series converges unconditionally if every rearrangement converges to the same sum. In finite-dimensional normed spaces, unconditional convergence is equivalent to absolute convergence. This property legitimizes regrouping binary-scale summands and separating blocks. The repository proves absolute convergence for the global signed and complex q-series, so their values do not depend on summation order.

Uniform approximation. Uniform approximation means convergence in supremum norm: $\sup_x |f_n(x) - f(x)| \to 0$. It preserves continuity and provides one error bound for all inputs. The centered finite splines approximate the bounded Fabius function uniformly with an explicit dyadic rate, and the Fourier-Legendre series of the up-function converges uniformly as well. The proofs use different mechanisms: probabilistic coupling for splines and coefficient bounds for polynomials.

Uniform convergence. A sequence f_n converges uniformly to f if for every $\varepsilon > 0$ one index works for all points in the domain. Uniform limits of continuous functions are continuous, and uniform convergence permits integration interchange on finite measure domains. It is stronger than pointwise convergence. The Fabius development repeatedly upgrades structural approximations to uniform ones so they can support computability and exact shape arguments.

Uniform distribution. The uniform distribution on $[a, b]$ has constant density $1/(b - a)$ on that interval. Its characteristic function is a phase times a sinc factor, and its moments are elementary rational expressions. Independent dyadically scaled uniforms are the building blocks of the Fabius random series. Centering removes the phases in the product transform and yields the even up-density.

Uniform norm. Uniform norm is another name for the supremum norm on a function space.

Convergence in this norm is uniform convergence. It is the metric used for the contraction fixed-point theorem, finite-spline error, and sharp derivative bounds in the Fabius development. On unbounded domains, boundedness or constant tails ensure the norm is finite.

Uniform random variable. A uniform random variable on $[a, b]$ assigns probability proportional to interval length. It can be realized as $a + (b - a)U$ with $U \sim \mathrm{Unif}[0, 1]$. Boundedness gives all moments and exponential moments. The Fabius law is assembled from independent uniform variables with geometrically shrinking coefficients, so finite partial sums have spline densities.

Uniform spline. A uniform spline has knots on an equally spaced grid. Cardinal B-splines are the standard examples. Although the original Fabius summands occur at different dyadic scales, a common refinement converts finite convolutions into uniform-grid splines with explicit coefficients. Centered uniform splines form the constructive approximation used in the computability theorem.

Unimodality. A function or density is unimodal if it increases up to a mode and decreases afterward, allowing nonstrict plateaus in the basic definition. Strict unimodality requires strict inequalities away from the unique mode. The up-function is strictly unimodal with mode zero. Consequently the Fabius density rises then falls, and the CDF changes from convexity to concavity at its unique midpoint inflection.

Uniqueness of the periodic correction. Suppose two continuous one-periodic functions inserted into the same Fabius endpoint main term both leave an error tending to zero. Their difference, evaluated along the unbounded lower-Lambert coordinate, tends to zero. Because every phase modulo one is approached by suitable endpoint sequences, periodicity and continuity force the difference to vanish identically. Thus the correction is uniquely determined, not merely one convenient Fourier representation.

22 V

Valuation. A discrete prime valuation $v_p(r)$ records the exponent of a prime p in a nonzero rational number. It is additive on products and satisfies a non-Archimedean inequality on sums. Valuations translate divisibility into arithmetic inequalities and identify reduced denominators. The Fabius arithmetic emphasizes v_2, but odd-prime valuations also underlie least-common-multiple denominator bounds.

Vandermonde determinant. For distinct nodes $x_0, \ldots, x_n$, the Vandermonde determinant is $\det(x_i^j)_{0 \le i,j \le n} = \prod_{i<j}(x_j - x_i)$, which is nonzero. It proves uniqueness of polynomial interpolation and invertibility of the evaluation map. On geometric nodes, its factors split into powers of two and q-Pochhammer products, explaining Gaussian interpolation weights in the dyadic formulas.

Vanishing moment. A signed measure, sequence, or function has vanishing moments through order $m-1$ if its integrals or sums against x^r vanish for $r<m$. Vanishing moments annihilate low-degree polynomials and improve approximation or decay. Complete Thue–Morse blocks have vanishing discrete moments up to an order determined by block length. This is the Prouhet cancellation used in Fabius finite formulas.

Variable substitution. Variable substitution transforms an integral or equation by a change of coordinates and includes the appropriate Jacobian. Affine substitutions connect F and up; logarithmic substitution converts dilation to translation; saddle substitution centers and scales the Bromwich contour. A flawed stationary substitution can leave an unaccounted linear term, so exact algebra and error propagation are essential.

Variance. The variance of a square-integrable random variable is $\mathrm{Var}(X)=\mathbb{E}[(X-\mathbb{E}X)^2]=\mathbb{E}[X^2]-(\mathbb{E}X)^2$. Variances add for independent variables. It is the second cumulant and sets the Gaussian scale in central approximations. For the dyadic uniform random series, the variance is an explicit geometric sum and is encoded among the centered moments.

Vertical Bromwich line. A vertical Bromwich line is the contour $\Re s=c$ used in inverse Laplace integration, oriented from $c-\mathrm{i}\infty$ to $c+\mathrm{i}\infty$. The constant c is chosen inside the transform's inversion region and to the right of relevant singularities. Saddle analysis may shift or deform this line after proving analyticity and decay. Orientation and the factor $1/(2\pi\mathrm{i})$ determine the sign and normalization.

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23 W

Weak convergence. Probability measures μ_n converge weakly to μ if $\int f\,\mathrm{d}\mu_n\to\int f\,\mathrm{d}\mu$ for every bounded continuous f. Characteristic-function convergence often proves weak convergence. Finite convolution laws of the truncated Fabius random series converge weakly to the infinite law because the random variables converge almost surely and remain uniformly bounded. Additional regularity yields convergence of densities or CDFs.

Weak-star convergence. Weak-star convergence of finite measures tests convergence against continuous compactly supported functions, viewing measures as dual objects. For probability measures on a compact set it closely matches weak convergence. The repository's early measure bridge uses weak-star convergence of convolution measures before identifying the smooth density and uniform CDF limit. The star distinguishes dual-space convergence from weak convergence inside the primal function space.

Weierstrass M-test. The Weierstrass M-test states that if $|f_n(x)|\le M_n$ for every x and $\sum M_n<\infty$, then $\sum f_n$ converges absolutely and uniformly. On compact subsets it proves normal convergence of holomorphic series. The repository uses summable majorants for Fourier modes, global binary summands, and transform logarithms. Uniform convergence then licenses continuity and termwise operations.

Weierstrass product. A Weierstrass product represents an entire function by factors encoding its zeros, supplemented when necessary by exponential convergence factors. The classical example is

$$\frac{\sin \pi z}{\pi z} = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right).$$

Substitution into $\widehat{\mathrm{up}}(z) = \prod_{j \geq 0} \operatorname{sinc}(\pi z/2^j)$ shows that a nonzero integer m is a transform zero of multiplicity $1 + v_2(m)$. This structure drives partition and sampling identities. A zero set determines an entire function only up to a zero-free factor unless normalization and growth conditions fix the ambiguity.

Weighted path expansion. A weighted path expansion unfolds a recurrence as a sum over directed paths through its dependency graph, multiplying transition weights along each path. It converts recursive definitions into finite closed sums indexed by compositions or subsets. Saddle coefficient and moment recurrences can be represented this way. In Lean, path expansions can make positivity, denominator structure, and coefficient dependence easier to prove than repeated symbolic substitution.

Weighted random sum. A weighted random sum is $\sum a_n X_n$. Its mean, variance, support, transforms, and convergence depend on both weights and component laws. Geometric weights give self-similarity and compact support when the variables are bounded. The Fabius law is the prototypical infinite weighted sum of independent uniforms, with base-two weights producing exact dyadic structure.

Well-founded recursion. Well-founded recursion defines a value using recursive calls on arguments smaller in a relation with no infinite descending chain. It generalizes primitive recursion on natural numbers. The exact dyadic Fabius evaluator removes the highest set bit at each call, so binary weight, or the numerator itself, supplies a decreasing termination measure. Induction over the same relation then proves correctness and complexity properties. This makes the algorithm mathematically finite even though the underlying Fabius function also has infinite-convolution and fixed-point definitions.

24 Z

Zero of an entire function. A zero of order m of an entire function F at z_0 satisfies $F(z) = (z - z_0)^m G(z)$ with $G(z_0) \neq 0$. Zeros are isolated unless the function is identically zero. The sinc product locates transform zeros at dyadically structured frequencies, with multiplicities determined by how many factors vanish. These zeros imply exact Fourier coefficient vanishings, partitions of unity, and odd-harmonic cosine expansions.

Zero run. A zero run is a consecutive block of zero digits or zero values. In binary recursions, long zero runs correspond to inactive dyadic scales and can be skipped by highest-set-bit algorithms. In coefficient sequences, a zero run may reflect parity, support, or finite-degree annihilation. The Fabius binary evaluator exploits bit structure rather than iterating through every zero position.

Zeta regularization. Zeta regularization assigns finite values to certain divergent products or sums by analytic continuation of a zeta-type function. It is conceptually related to, but distinct from, the Hadamard finite part used in the Fabius Mellin analysis. The repository's Gamma–zeta coefficients arise from ordinary meromorphic continuation and residue calculus; no arbitrary assignment of a divergent original series is intended. The term is included because zeta-based regularization language appears in discussions of finite parts and must be distinguished carefully.

A Notation and formalization crosswalk

The declarations below are representative entry points rather than an exhaustive API. A row fixes the normalization used in this glossary and points to the Lean declaration whose type and hypotheses govern formal use.

Object	Meaning and normalization	Lean declaration(s) and module
F	Bounded CDF-style Fabius function, constant outside $[0,1]$.	`BoundedFabius`, `fabiusReal`, `IsFabius`; `Basic.lean`.
up	Even compactly supported fold $\mathrm{up}(x) = F(1-\lvert x\rvert)$.	`rvachevUp`; `Basic.lean`.
$\mathcal{F}$	Signed global extension, not clamped outside $[0,1]$.	`extendedFabius`; `Basic.lean`.
$\mathrm{wt}_2(n)$, ε_n	Binary digit sum and Thue–Morse sign $(-1)^{\mathrm{wt}_2(n)}$.	`binaryWeight`, `thueMorseSign`; `Arithmetic.lean`.
c_n	Full moment $\int_{\mathbb{R}} t^{2n}\,\mathrm{up}(t)\ \mathrm{d}t$.	`moment`; `Arithmetic.lean`.
d_n	$d_0 = 1$ and $d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\ \mathrm{d}t$ for $n \geq 1$.	`halfMoment`; `Arithmetic.lean`.
F_n, G_n	Integer numerators of c_n, d_n in the displayed canonical normalizations; F_n is not the function F.	`momentNumerator`, `halfMomentNumerator`; `Arithmetic.lean`.
R_n	Reshetnikov normalization; a positive odd integer for $n \geq 1$.	`reshetnikov`; `Arithmetic.lean`.
D_n	LCM of reduced denominators of $\mathrm{up}(a/2^n)$ on the positive odd level-n grid; equal to the corresponding bounded-F LCM.	`dyadicDenominator`, `fabiusDyadicDenominator`; `Arithmetic.lean`.
W_{-1}	Lower real branch of Lambert W.	`lowerLambertW`; `LowerLambertW.lean`.
$\lambda(x)$	Lower-Lambert phase/reference saddle coordinate, with $\lambda 2^{-\lambda} = x$.	`fabiusLambertPhase`; `FabiusLambertSaddle.lean`.
Ψ	Centered smooth one-periodic correction; distinguished from its uncentered precursor.	`negativeLaplacePsi`, `negativeLaplacePeriodicCorrection`; `PeriodicCorrection.lean`, `NegativeLaplace.lean`.
A_j	One-periodic logarithmic coefficients in the all-orders endpoint expansion.	`fabiusSaddleLogCoefficient`; `FabiusSaddleExpansionCoefficients.lean`.

B Coverage map

The headwords cover the seven main mathematical layers that recur in the source tree:

1. the three Fabius normalizations, their functional-differential equations, shape, support, smoothness, fixed-point construction, and probability representation;
2. finite and infinite convolution, centered splines, weak convergence, Fourier transform, sinc products, Poisson summation, sampling, and partitions of unity;
3. moments, Bernoulli recurrences, exact dyadic evaluation, valuations, integer numerators, denominator bounds, and fast binary recursion;
4. Thue–Morse signs, Prouhet cancellation, geometric interpolation, half-base q-calculus, Gaussian binomial identities, complex shifts, and Toeplitz limits;
5. negative-Laplace products, Bose kernels, Mellin finite parts, Gamma–zeta Fourier modes, periodic corrections, Lambert W, and logarithmic delay structure;
6. Bromwich inversion, saddle points, local phase jets, major/minor arcs, tail estimates, first and second corrections, and the full all-orders expansion;
7. Legendre polynomials and least squares, effective flatness, dyadic names, primitive recursion, and Grzegorczyk computability.

C Source note

The source directory examined was

`https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction`.

The visual model was the LaTeX article at

`https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`.

The inventory was normalized at the level of mathematical concepts: repeated occurrences and plural forms do not create entries, bare spelling aliases are normally consolidated, and a role-specific alias is retained only when its definition makes that relationship explicit. Specialized compounds and repository-defined mathematical objects receive separate headwords.